

NORTH CAROLINA STATE UNIVERSITY



TEECA MANUFACTURING COMPETITION

DICE VAULT

ATLANTIC SANDS HOTEL & CONFERENCE CENTER

NOVEMBER 2-4 2023

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IMPORTANT SAFETY INFORMATION

These symbols are a safety alerts indicating potential hazards.
They are used to draw attention to critical information regarding the safety of
individuals in a given area or situation.

Please take immediate precautions and follow safety procedures as advised.

Your well-being is our top priority. Stay vigilant and act responsibly.

Remember, safety first!

Always wear appropriate eye and ear protection.



This symbol indicates an extremely hazardous situation. Failure to take immediate action and avoid this danger **will lead to severe injury or loss of life.**



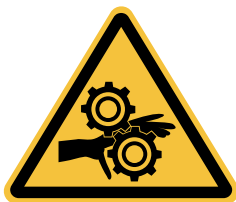
This symbol indicates an extremely hazardous situation. Failure to take immediate action and avoid this danger **may lead to severe injury or loss of life.**



This symbol indicates an extremely hazardous situation. Failure to take immediate action and avoid this danger **may lead to minor or moderate injury.**



This symbol indicates a cut or bisect hazard. Avoid contact with cutting tools during operation.



This symbol indicates an entanglement hazard. Avoid pinch point contact with machinery in motion during operations



This symbol indicates a fire or burn hazard. Avoid contact with hot components during machinery operation.

TEECA ADVISORS

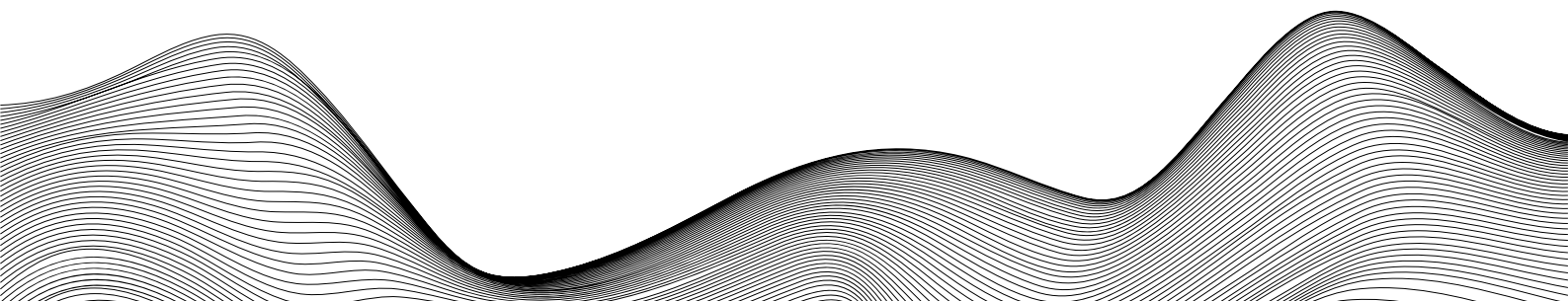
Steven Miller, EdD (Education Leadership & Policy Studies w/Adult & Higher Education)

Sam Munzer, Graduate Student (PhD in Learning and Teaching in STEM: Engineering and Technology Education concentration)

Joseph Kaskel, Graduate Student (PhD in Learning and Teaching in STEM: Engineering and Technology Education concentration)

Erik Schettig, Graduate Student (PhD Learning and Teaching in STEM: Engineering and Technology Education concentration)

Parks Newby, MEd (Industrial Arts Education - Minor in Industrial Engineering - Ergonomics)



TEAM ROSTER

Sean Ryan, Undergraduate (Technology, Engineering, and Design Education)
Led the CNC-focused operations, including tooling creation, documentation management, and overall team coordination leader.

Laina Hosey, Undergraduate (Technology, Engineering, and Design Education)

Led the design efforts for the foam cutter project, overseeing the development of the cutting mechanism and ensuring precise placement of the NCSU logo.

Wyatt Williams, Undergraduate (Technology, Engineering, and Design Education)

Led the development of the shooting board project, focusing on designing a tool for creating precise chamfers in woodworking projects.

Elliot Umbarger, Undergraduate (Technology, Engineering, and Design Education - Licensure)

Played a pivotal role in assuring the quality of wooden squares, overseeing inspections and implementing quality control measures throughout the production process.

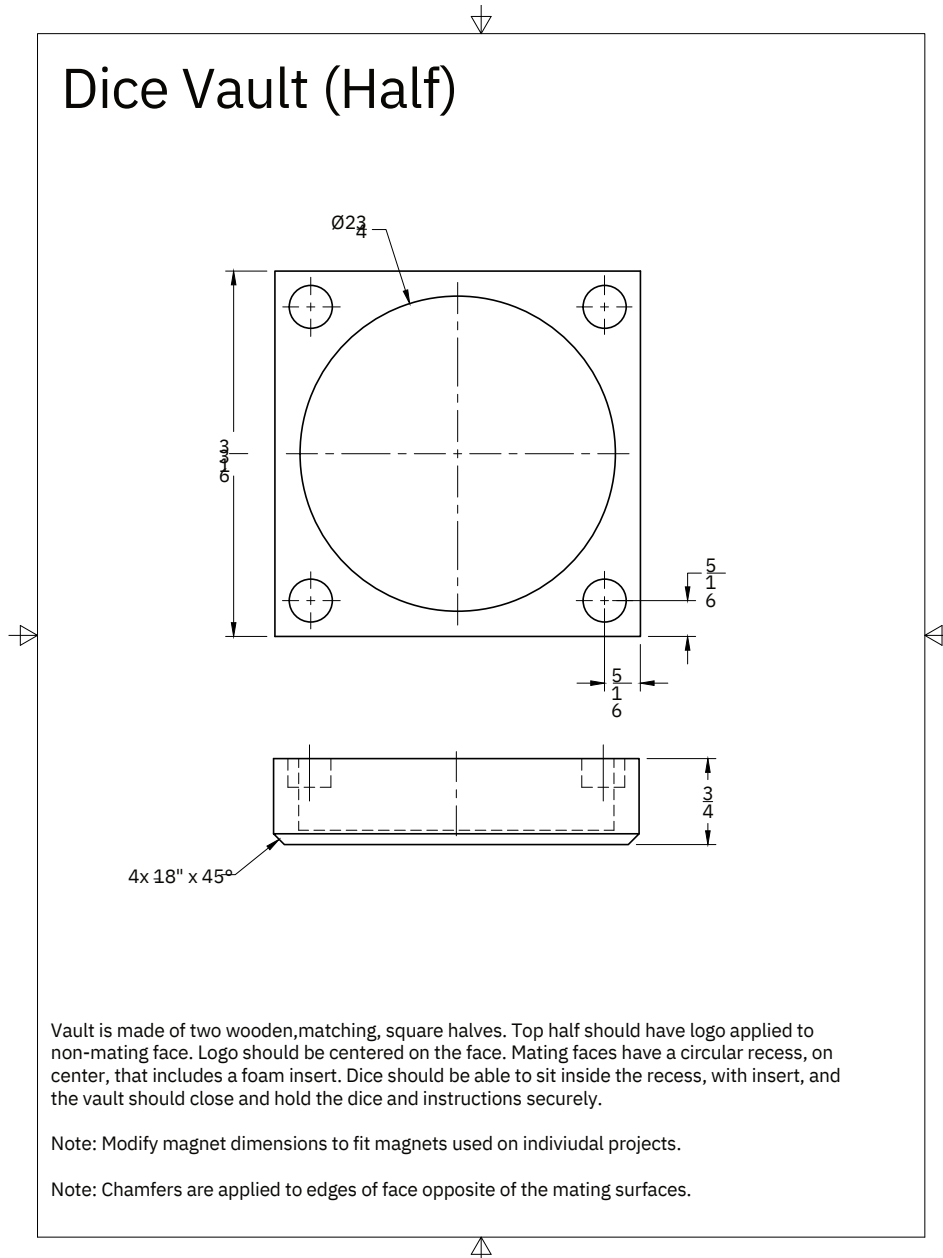
Lucas Ciscato, Undergraduate (Technology, Engineering, and Design Education - Licensure)

Collaborated closely with the Design Team Lead to ensure the successful development of the cutting mechanism and precise placement of the NCSU logo in the foam cutter project.

Luke Starkey Undergraduate (Technology, Engineering, and Design Education)

Played a crucial role in optimizing various aspects of the process, with a focus on ensuring the quality of both wooden squares and the chamfer shooting board.

CRITERIA



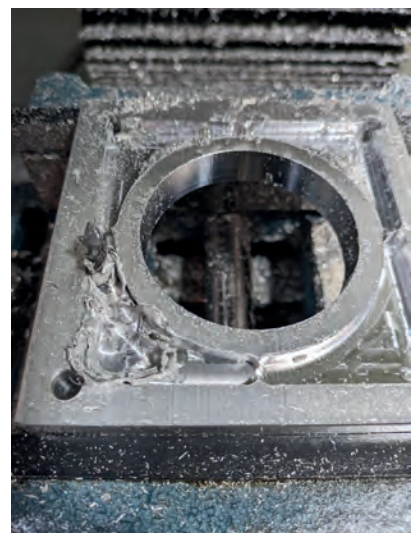
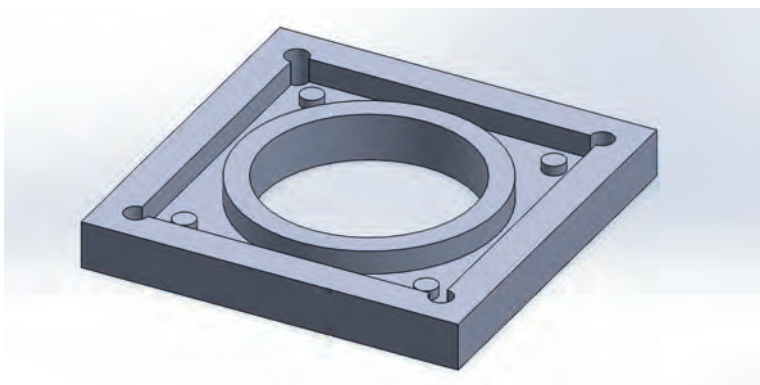
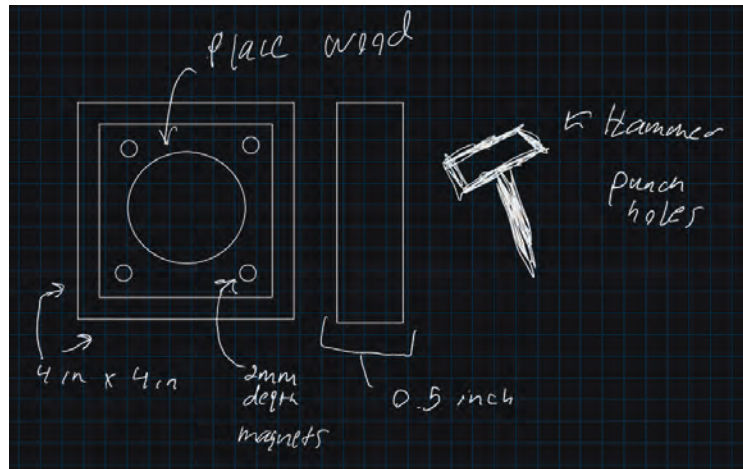
The evaluation criteria for the dice vault challenge encompass team performance, safety measures, craftsmanship in tooling design, documentation of design processes, and the overall quality of the final product. Problem-solving skills play a vital role, particularly in the creation of tooling and inspection gauges, encouraging teams to explore innovative fabrication techniques for improved efficiency and consistency. The utilization of advanced equipment like CNC machines, lasers, and automated systems was encouraged and utilized by NCSU, with an emphasis on providing a valuable learning experience for students. Additionally, thorough documentation and justification of material selection, equipment usage, and external partnerships are integral components of the presentation.

TOOLING DESIGN AND DEVELOPMENT

THE MAGNET HOLE STAMP

Magnet hole creation tool created out of 6061 aluminum. The aluminum circular boss aligns the wooden work piece for correct given holes. A vice is utilized to compress the magnet holes into the wood.

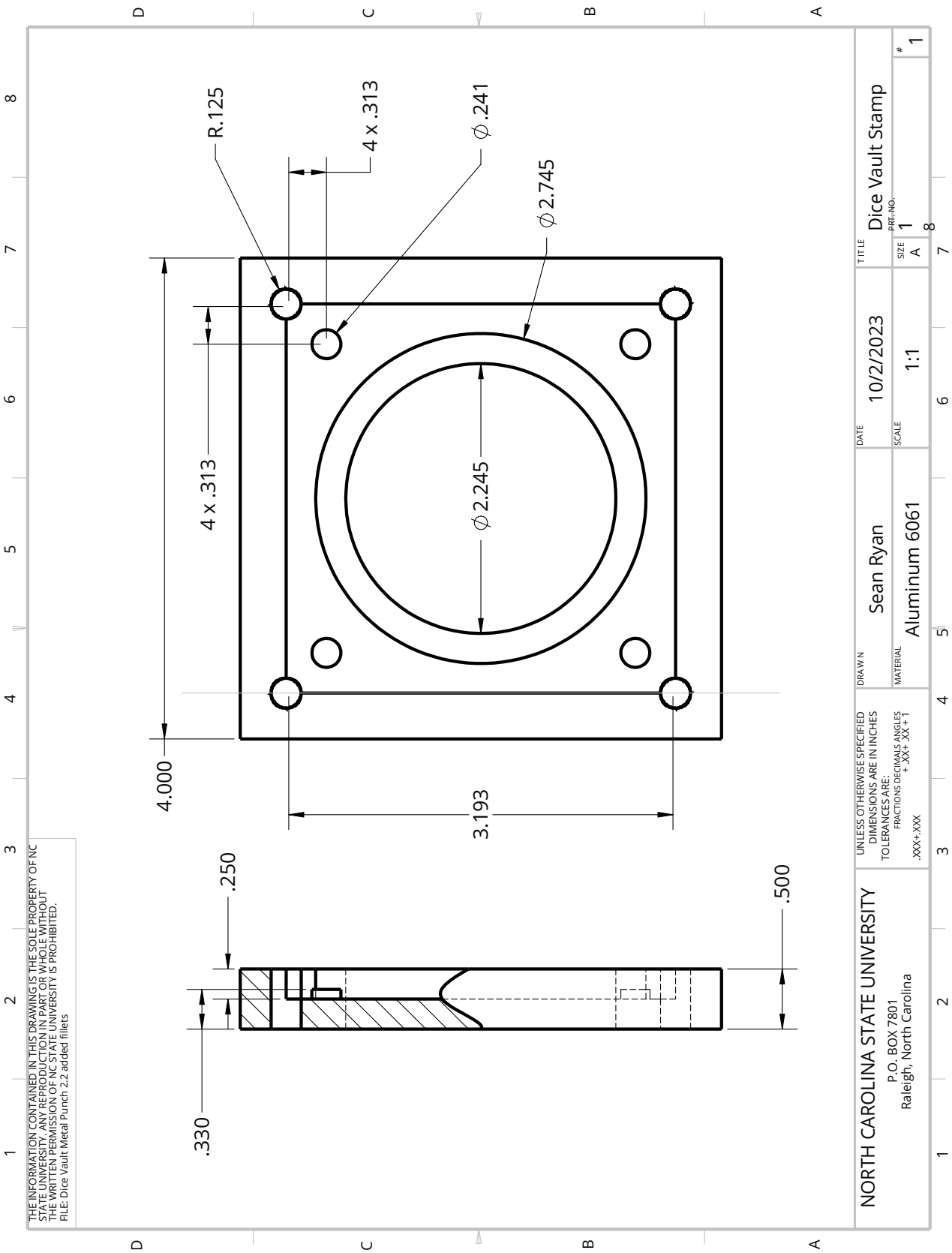
hand drawn sketch done on touch screen



Three bits were sacrificed in the name of CNC tech education. RIP.

TOOLING DESIGN AND DEVELOPMENT

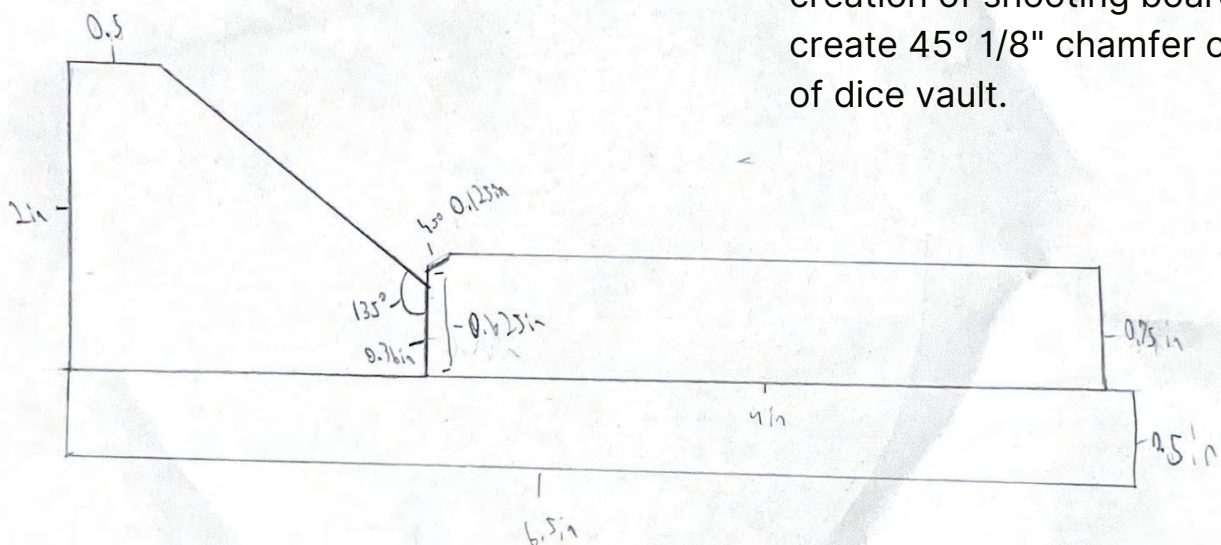
THE MAGNET HOLE STAMP



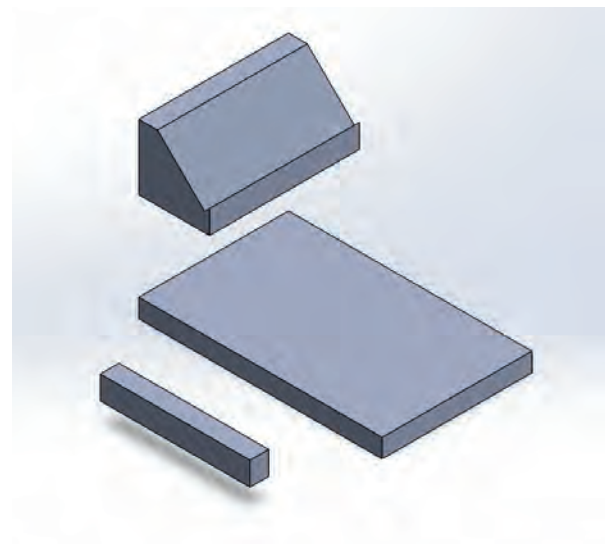
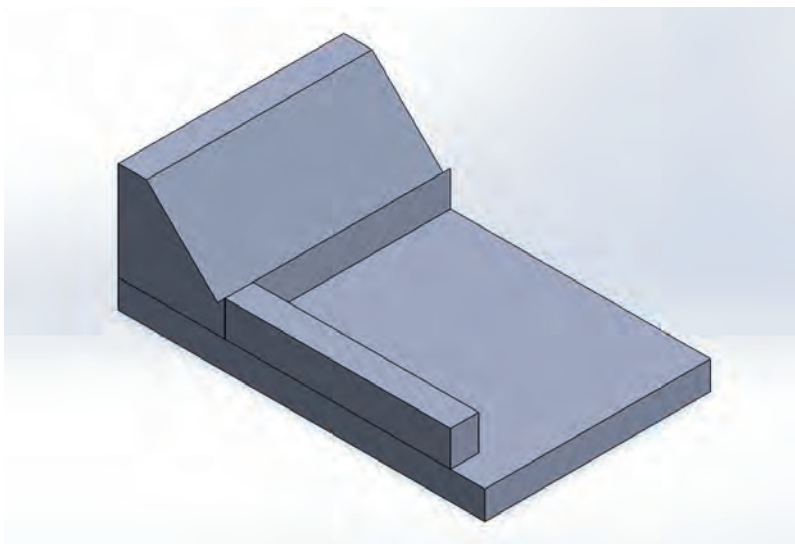
TOOLING - DESIGN AND DEVELOPMENT SHOOTING BOARD AND CHAMFER CREATION



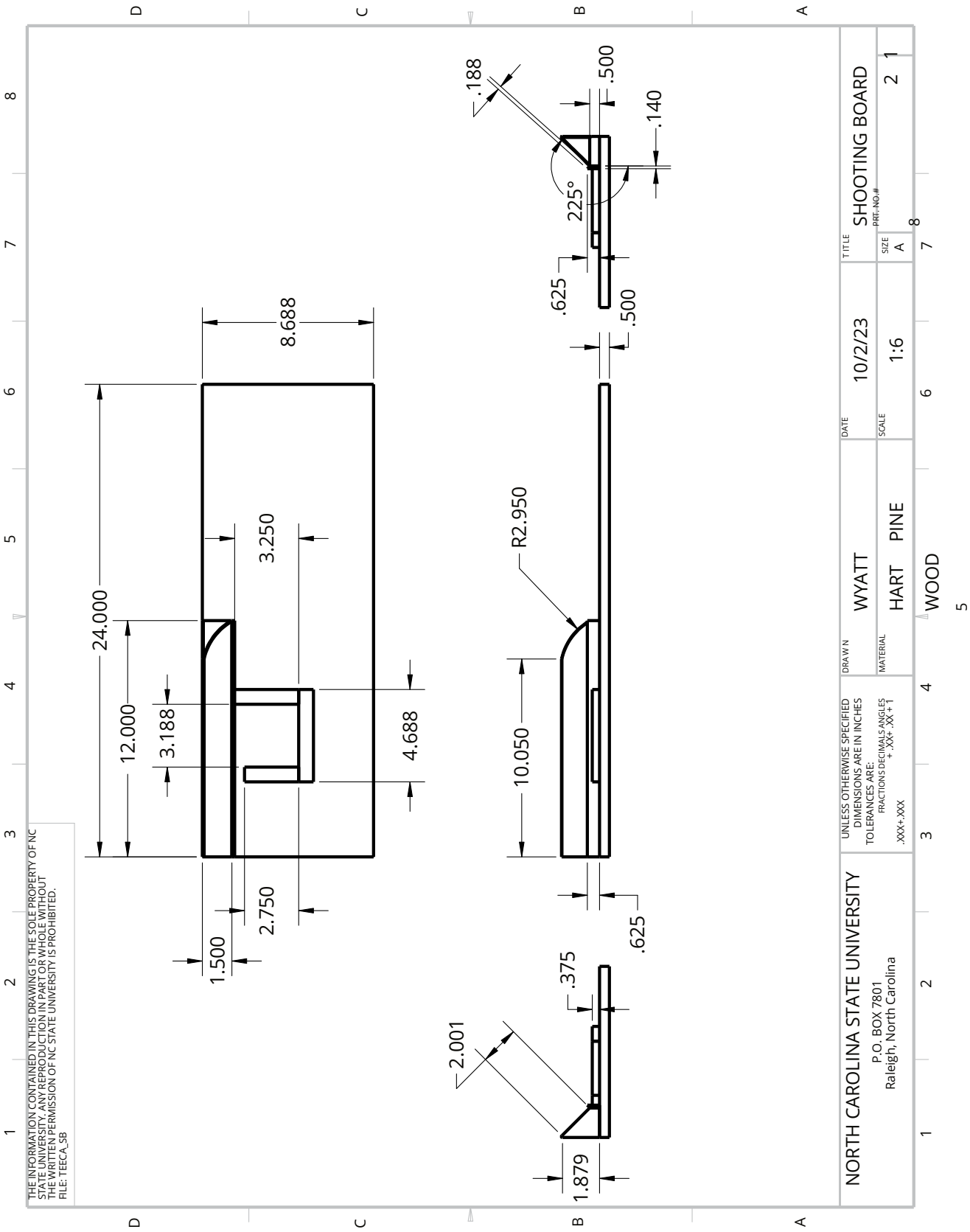
Hand drawn sketching for shooting board



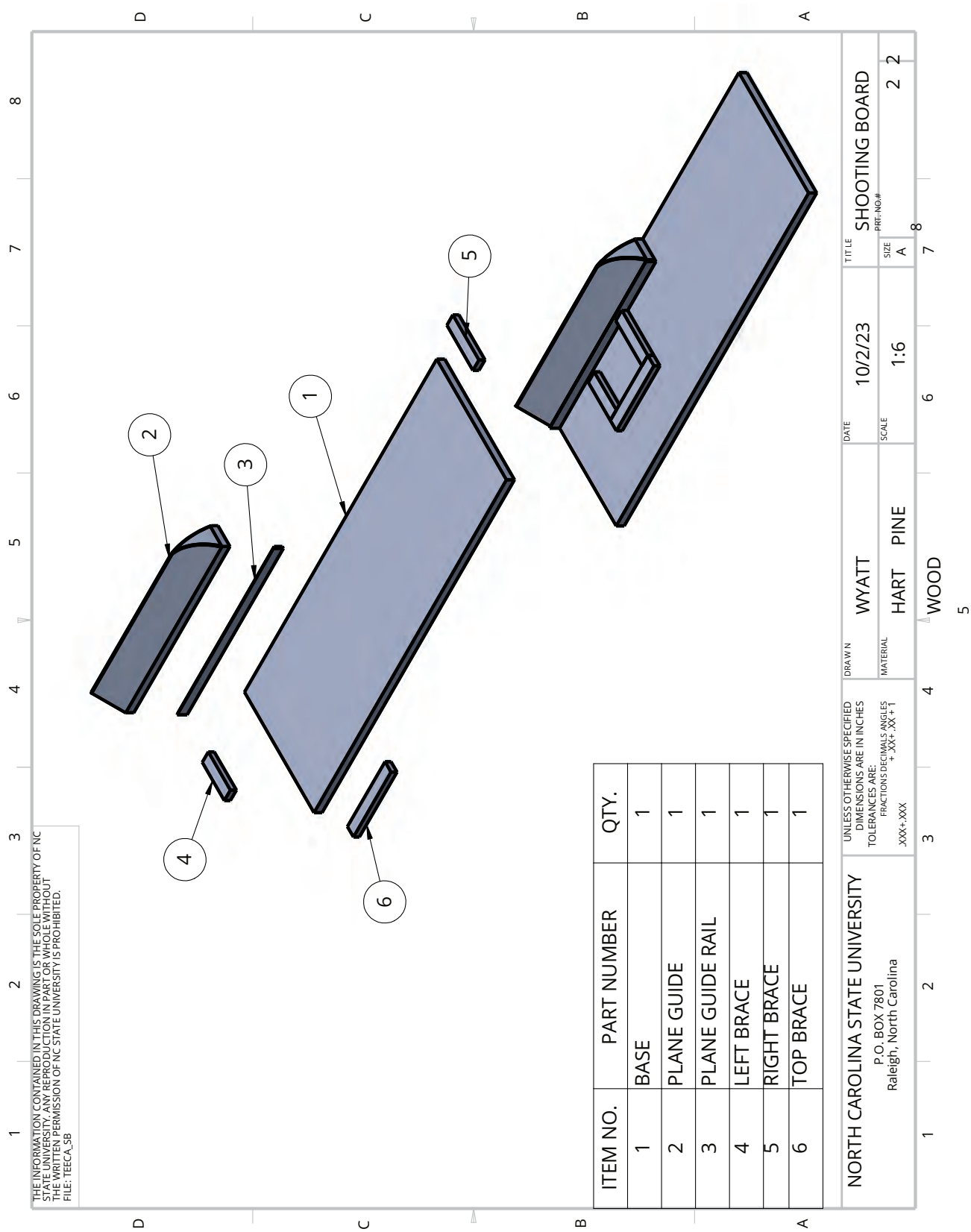
Solidworks was utilized in the creation of shooting boards to create 45° 1/8" chamfer on walls of dice vault.



TOOLING - DESIGN AND DEVELOPMENT
SHOOTING BOARD AND CHAMFER CREATION

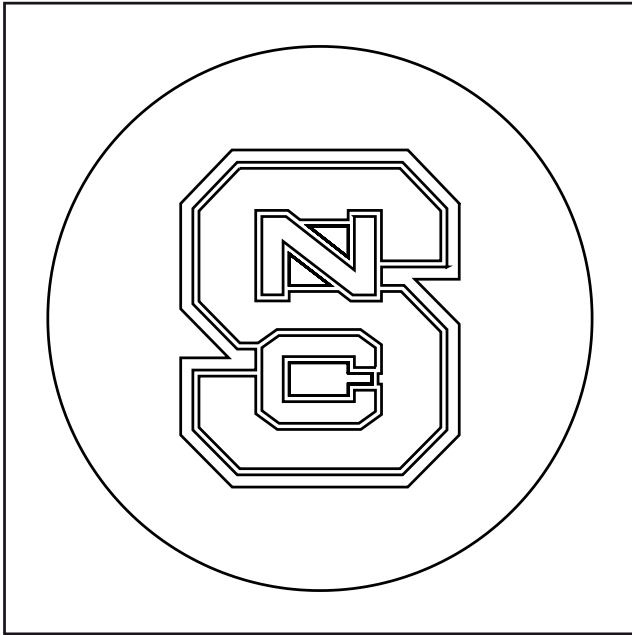


TOOLING - DESIGN AND DEVELOPMENT
SHOOTING BOARD AND CHAMFER CREATION



TOOLING - DESIGN AND DEVELOPMENT

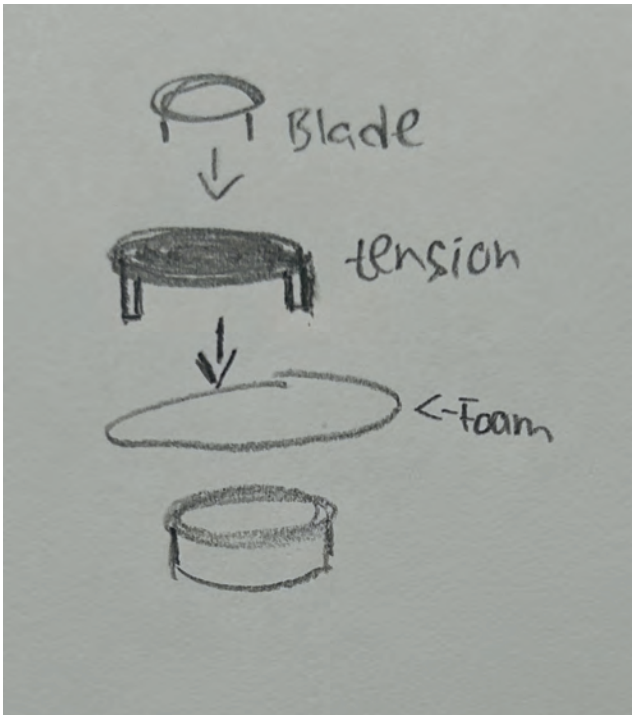
LOGO DESIGN & FOAM CUTTER



Tool Considerations Notes

Adobe Illustrator is an effective and easy way to import, edit, and export vector graphics. SVGs are an ideal file type as they can be changed to any size and maintain quality, as it is based on geometric equations as opposed to pixels. Our Thunder laser cutter also is compatible with SVGs and .Ai files. Adobe Illustrator allows us to see and edit the colors and width of the lines, as well as convert between CMYK and RGB color modes, allowing the design to be modified based on required inputs for desired outputs of our laser cutter.

This logo is one of our many NCSU logo designs, and fits well on a square lid. The design is sized so one axis is not overtly longer than the other, such as the full name logo, meaning that the design can be big enough to see and not leave large blank spaces unevenly on the lid. Saw blade was utilized over other cutting methods because the cutting point would be a complete 360 degrees efficiently compared to one cut point being turned over and over.



1 2 3 4 5 6 7 8

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FILE: Foam Cutting jig V2

D

C

B

A

ITEM NO. PART NUMBER MATERIAL QTY.

1	BLADE HOLDER	PLA	1
2	TENSION	PLA	1
3	BASE BLOCK	PLA	1

SECTION A-A
SCALE 1:3

1

2

3

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ARE:
+ .XX+ .XX + 1
.XXX+ .XXX

NORTH CAROLINA STATE UNIVERSITY
P.O. BOX 7801
Raleigh, North Carolina

DATE 11/2/2023

TITLE LAINA HOSEY

SCALE 1:3

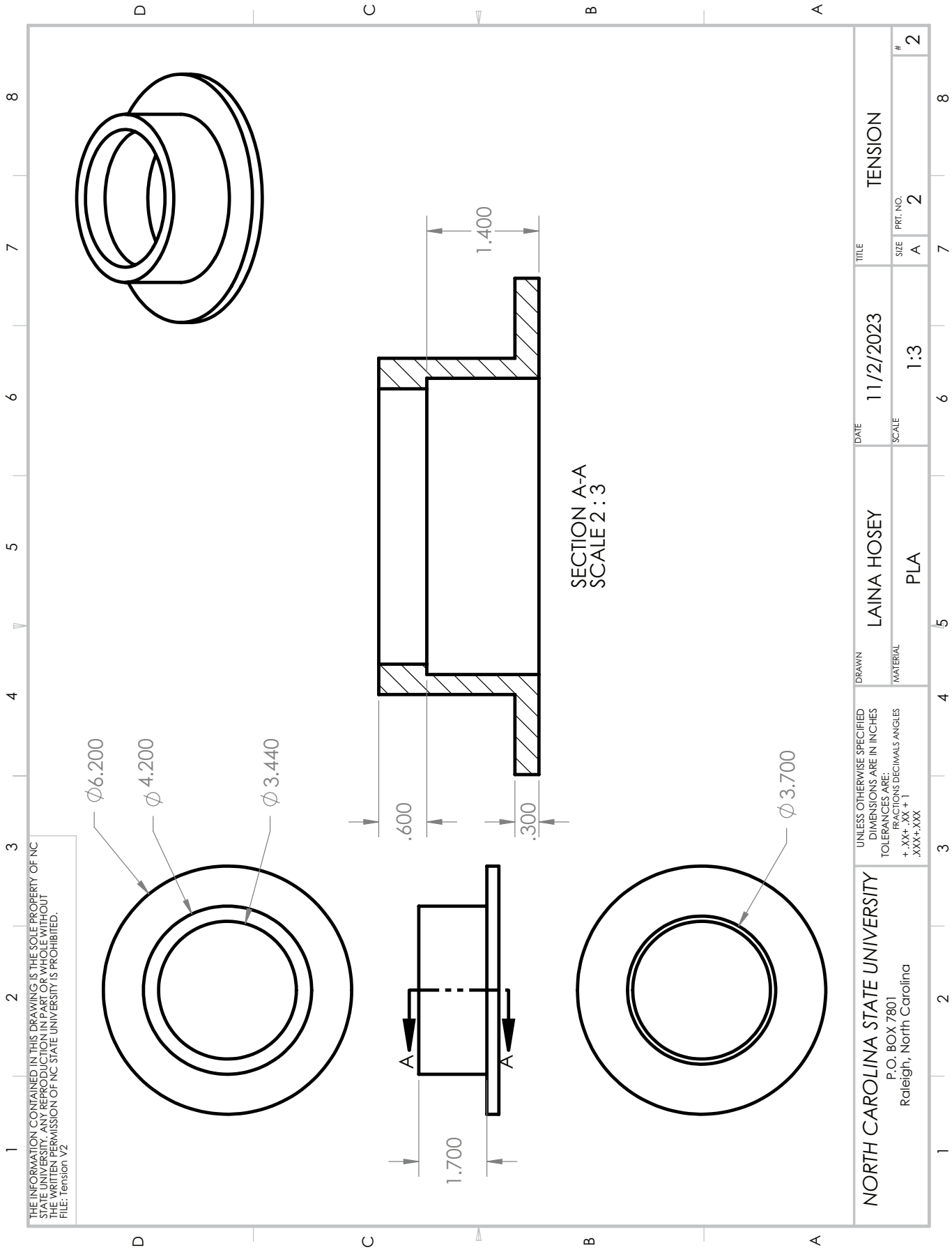
SIZE A

PRT. NO. 1-3

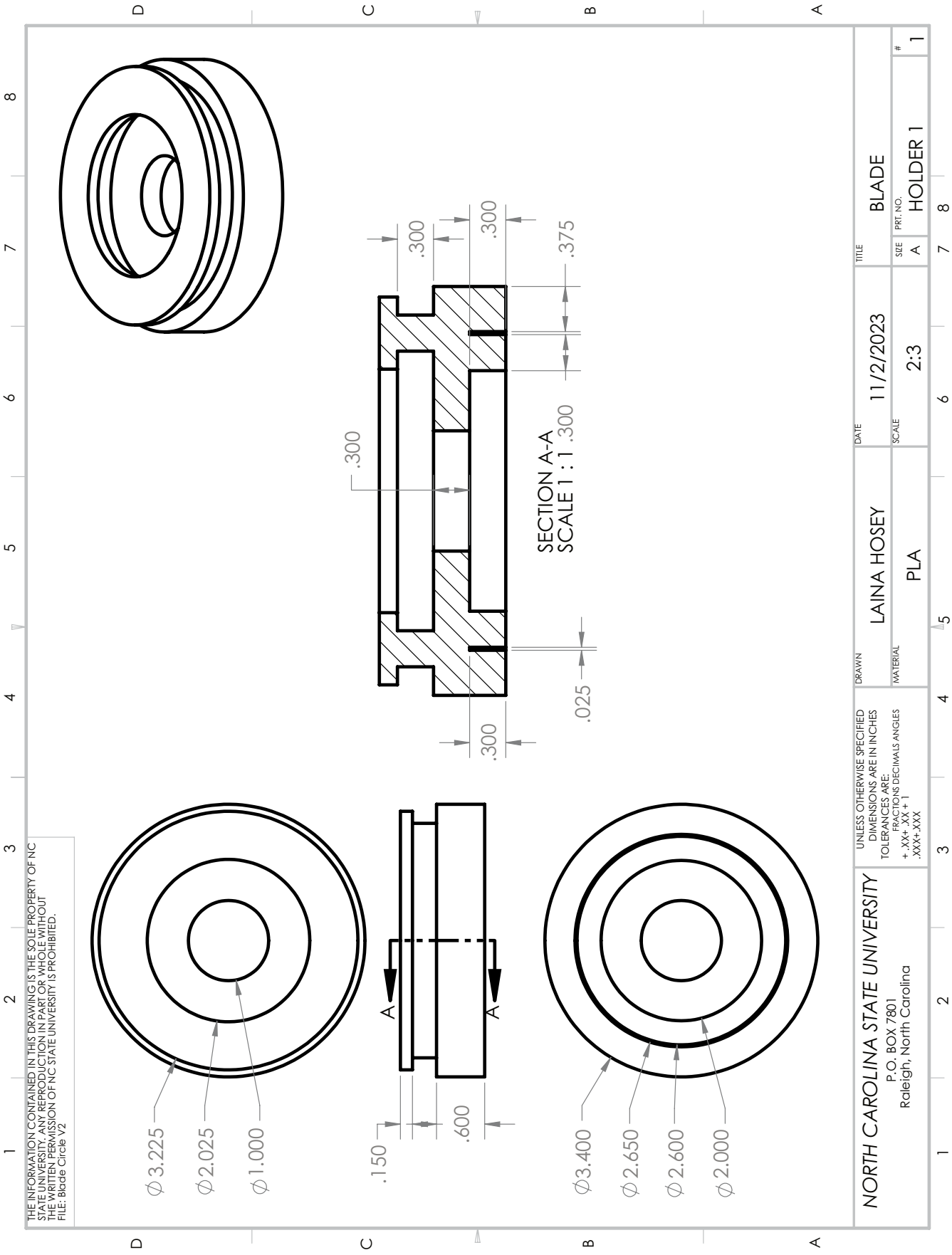
1

1 2 3 4 5 6 7 8

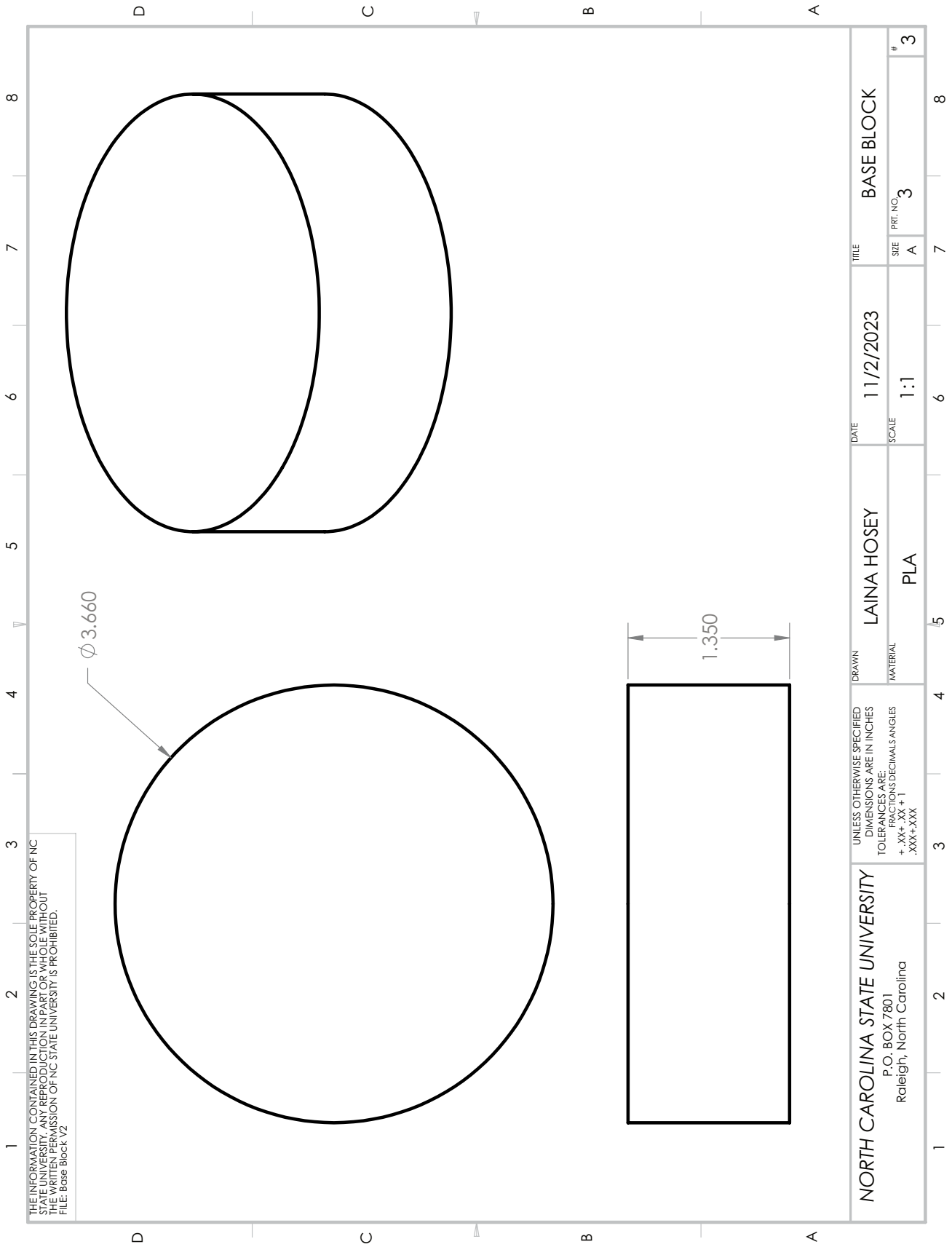
TOOLING - DESIGN AND DEVELOPMENT
LOGO DESIGN & FOAM CUTTER



TOOLING - DESIGN AND DEVELOPMENT
LOGO DESIGN & FOAM CUTTER

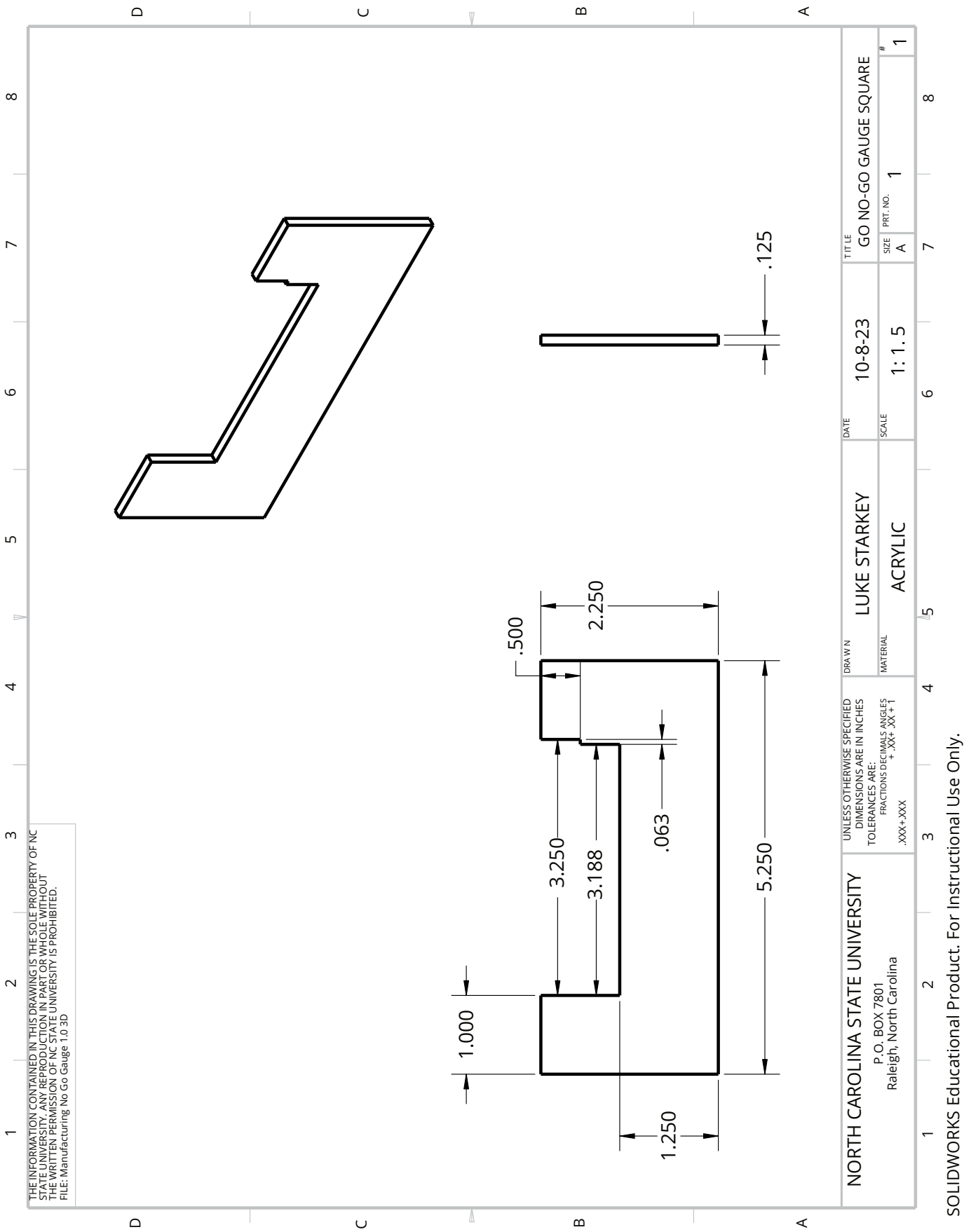


TOOLING - DESIGN AND DEVELOPMENT
LOGO DESIGN & FOAM CUTTER

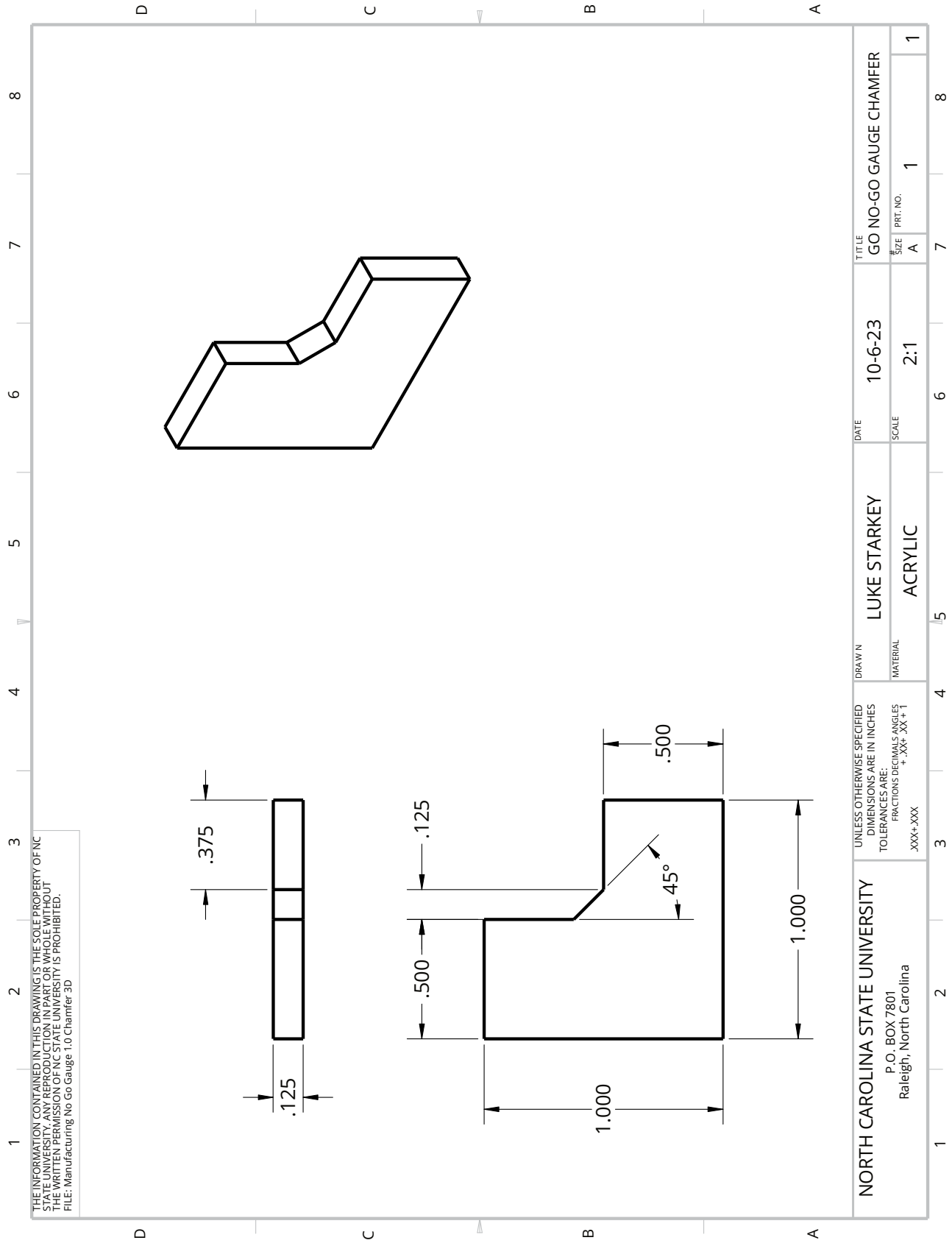


TOOLING - DESIGN AND DEVELOPMENT

GO NO-GO GAUGES



TOOLING - DESIGN AND DEVELOPMENT
GO NO-GO GAUGES

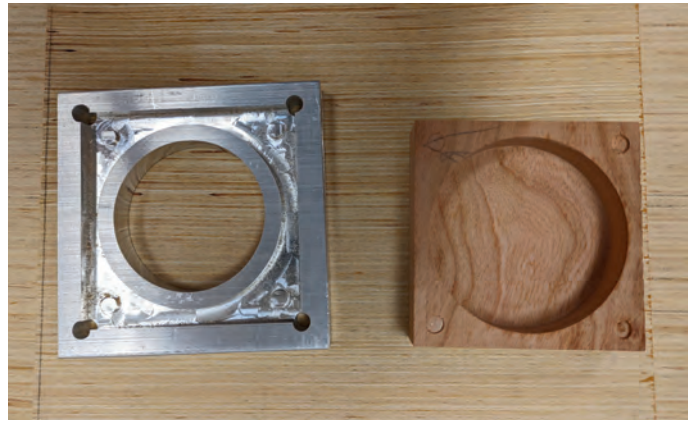


SOLIDWORKS Educational Product. For Instructional Use Only.

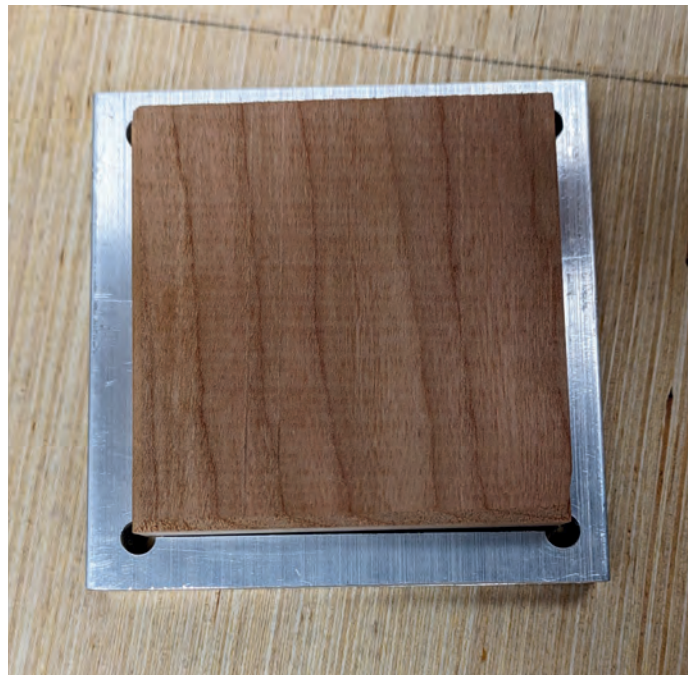
TOOLING RATIONALE- MAGNET HOLE STAMP

Tool Considerations Notes:

The first tool created was a magnet hole stamp. While the initial designs were focused on utilizing a 3D print to orient a drill. It was determined that the process could be stream-lined by making a metal stamp.



Adjustability/Alignment: The center hole ensures that each wood square is center, doesn't move, and magnet holes are oriented correctly.



Safety: A vice is utilized to press the holes into the wood. Care must be taken to ensure that fingers do not become pinched during use. Gloves should also be worn when removing wood in order to prevent splinters or lacerations.

Ensures Accuracy: The CNC processed aluminum ensures a high level of accuracy. A tolerance of $1/16"$ was included to account for fluctuations in wood humidity. The center piece has a boss to ensure the wood is centered properly. The accuracy of this piece was the main reason for moving forward in this direction and ensures that our magnet holes are lined up with each dice vault. Most important compared to a drill press, this process is much quicker.



TOOLING RATIONALE- SHOOTING BOARD

We aimed to construct a tool that could consistently make four $\frac{1}{8}$ " 45° chamfers. Upon conducting research we settled on developing a custom shooting board to hold the piece and cut the chamfers using a plane.

Chip Clearance: shooting board designed to be easily cleaned in preparation of switching out wood squares.

Tolerance: In order to consistently create identical chamfers a majority of the focus to begin was creating a guide for the plane that would cut at 45° angle and could only cut deep enough for the chamfer to be $\frac{1}{8}$ ". This led to the design and construction of the plane guide which includes one larger 45° wall that properly angles the plane and a smaller 45° wall that acts as a stop limiting the depth of the cut.

Adjustability: For further quality assurance, we made the option to add or subtract paper shims to ever so slightly adjust the depth of the cut which affects the chamfer width. These adjustments can be made after the initial cut is made and the accuracy is checked using a Go No-Go of acrylic with the negative of the chamfer.

Safety: To ensure safety all cuts are made moving away from the user's body, the piece itself is also held in place by 3 braces so the user's hands are not in the cutting path. For further hand safety, the plane guide fileted to allow for a smooth user experience.



TOOLING RATIONALE- FOAM CUTTER

The graphic design team focused on creating a foam cutter. To ensure functionality, accuracy, and safety, they considered different possibilities on their approach to 1) cutting the foam, 2) the housing of the tool, and 3) the type of blade used. The indentation hardness of the foam was suitable to be cut by a the saw blade and also ideal for sublimation for the NCSU logo graphic.

Functionality: Initial considerations included a hole punch and a rotational blade design. After careful deliberation, the design team determined that due to the foam indentation hardness, it would be difficult to punch a hole. Instead, the design team decided to pursue a rotational blade design. Additionally essential aspect of the functionality was tension. The housing had to create tension to properly cut the foam.

Accuracy: The housing was an essential part of ensuring accuracy and repeatability. There were discussion between using either constructing the housing out of wood or 3D printing it. Since accuracy is an important aspect of the result, the design team settled with 3D printing. They also made sure that the housing created enough tension that would allow for a clean and consistent cut. Another important aspect in consideration was the type of blade used. The design team debated between implementing a single exacto blade or a saw blade that circumscribed the foam piece. They decided that the single-blade solution would result in inconsistencies and therefore pursued the latter.

Safety: The use of sharp object obviously poses potential hazards, so safety was a vital element. The blade design as well as the housing were at the at most importance for this detail. The housing was designed to keep the blade away from the operator. It is completely enclosed while in use. The final reason for why they decided on a saw blade rather than the exacto blade was because of the exacto blade's hazardous implications.



TOOLING RATIONALE- GO NO-GO GAUGE

Overview

The acrylic go no-go gauge is a crucial tool designed for quality control. Made in the Poe Fabrication space with the Thunder Laser Cutter System. These tools serve two primary purposes: one ensures the precise dimensions of the square feature (3 3/16" with a tolerance of 1/16") and the other verifies the presence of a 1/8" 45-degree chamfer on the squares. The rationale for implementing this tool can be simplified as a fast and accurate tolerance check.



Accuracy

Precision in Measurement: The acrylic go no-go gauges are essential to maintain accuracy in the fabrication of our dice boxes. They enable users to verify that the square and the chamfer dimensions are within a 1/16" tolerance. This level of precision ensures the exactness required for high-quality fabrications.

Functionality

Time Efficiency: The use of the no-go gauge streamlines the inspection process, reducing the time required for manual measurements. This efficiency not only improves productivity but also minimizes the likelihood of human errors.

Compatibility: Its ease of use ensures that all team members can utilize it effectively. It aligns with our team's workflow by saving us time and improving efficiency. The go no-go gauges are easy to use, and efficient aligning with the tooling design objective of our team, aimed at creating a streamlined process.



Safety

By ensuring that components meet design specifications, the go no-go gauge helps prevent rework or the use of substandard components, reducing the potential for safety hazards in the fabrication process. The quality assurance provided by the go no-go reduces variability in the product, preventing potentially hazardous situations. The hazardous situations could arise from tooling errors caused by components that are out of tolerance.

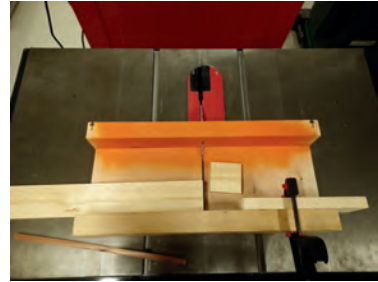
The practice of checking tolerances as you go not only improves overall accuracy but reduces the possibility of reworking products at the end. In a competition where time is limited, any last minute work can become dangerous. Having to rework products could cause fabricators to get into a rush which greatly increases the likelihood of an injury.



TOOLING RATIONALE

Tool Considerations Notes:

Table saw was used as a starting point in order to cut wood to correct width (3 3/16" square).



Tool Consideration Notes:

The Tormach CNC 1100m was utilized because it could create a precise, repeatable, and superior surface finish on the floor of the pocket compared to a drill press bit.



TOOLING RATIONALE

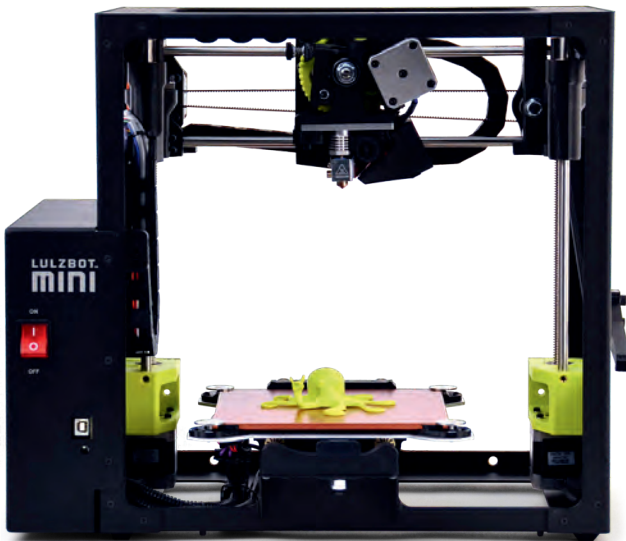
Tool Consideration Notes:

The WAZER (Water jet cutter) was utilized for its accuracy and precision in order to make initial prototype metal templates for a drill line up fixture. Not utilized for final tooling.



Tool Consideration Notes:

Foam cutting jig has a unique shape that demanded a printer with precise detailing capabilities, and the Lulzbot Mini proved to be the ideal candidate.



Tool Consideration Notes:

Laser cutter ideal for cutting Go No-Go Gauges with precision. Also utilize to mark center point to place sticker for dice vault tops.



TOOLING RATIONALE

Tool Considerations Notes:

Miter Saw was utilized for initial length cuts to bring wood to a working length on the saw stop.



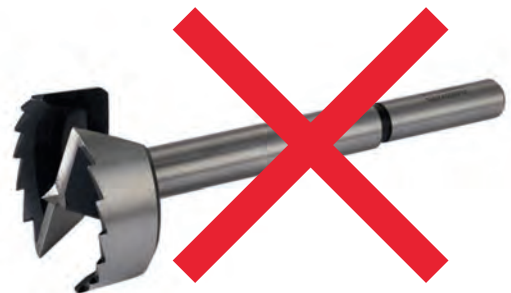
Tool Considerations Notes:

Jointer was utilized to smooth out rough sides of wood. Without the use of a jointer, it can be difficult to ensure that all sides of the wood are equal in length and perpendicular before moving to the planer and table saw



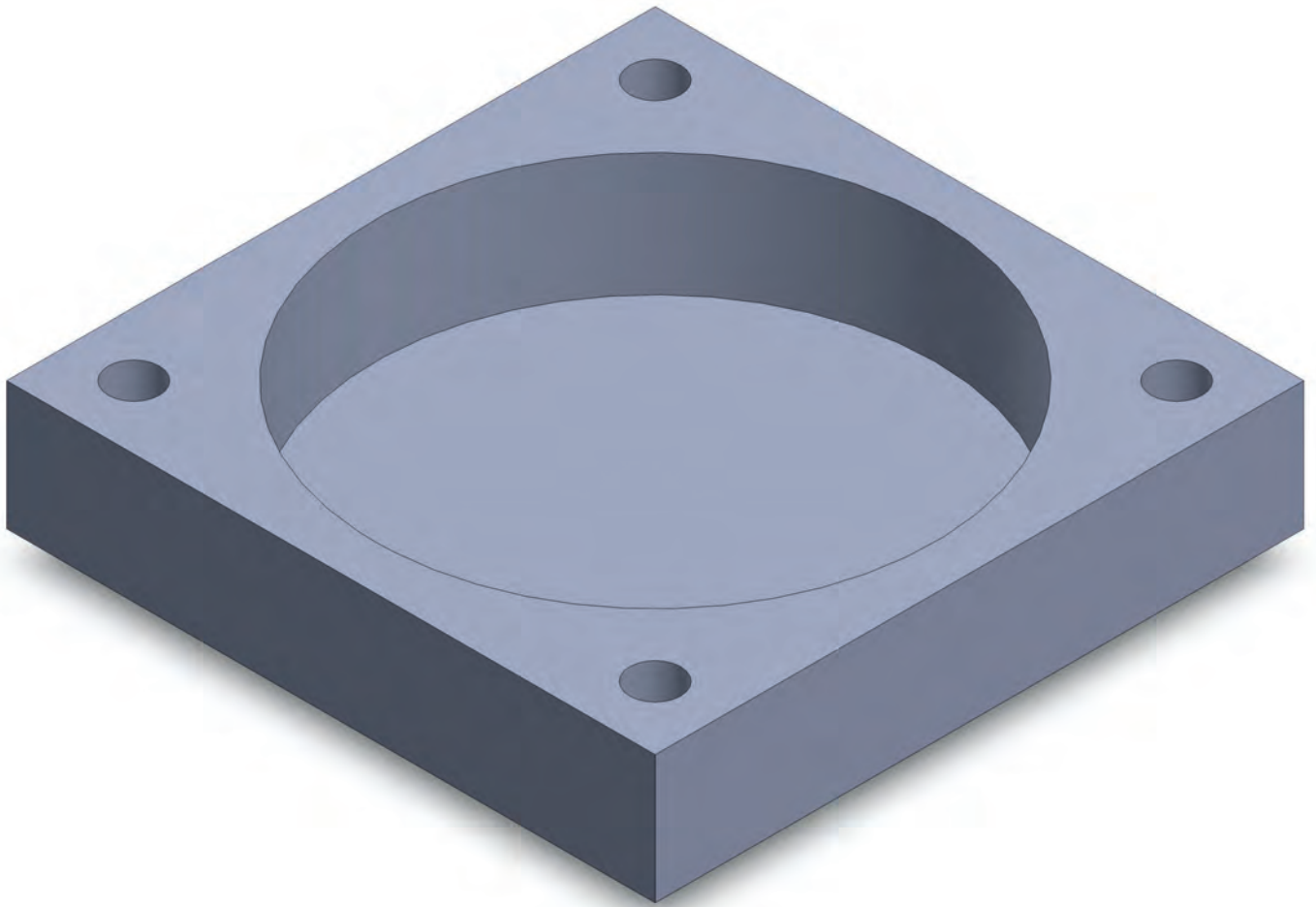
Tool Considerations Notes:

Forstner 2-3/4 in diameter bit. The dice vault pocket portion leaves so little wood that a Forstner bit center would drill an unsightly and unwanted hole through the dice vault.

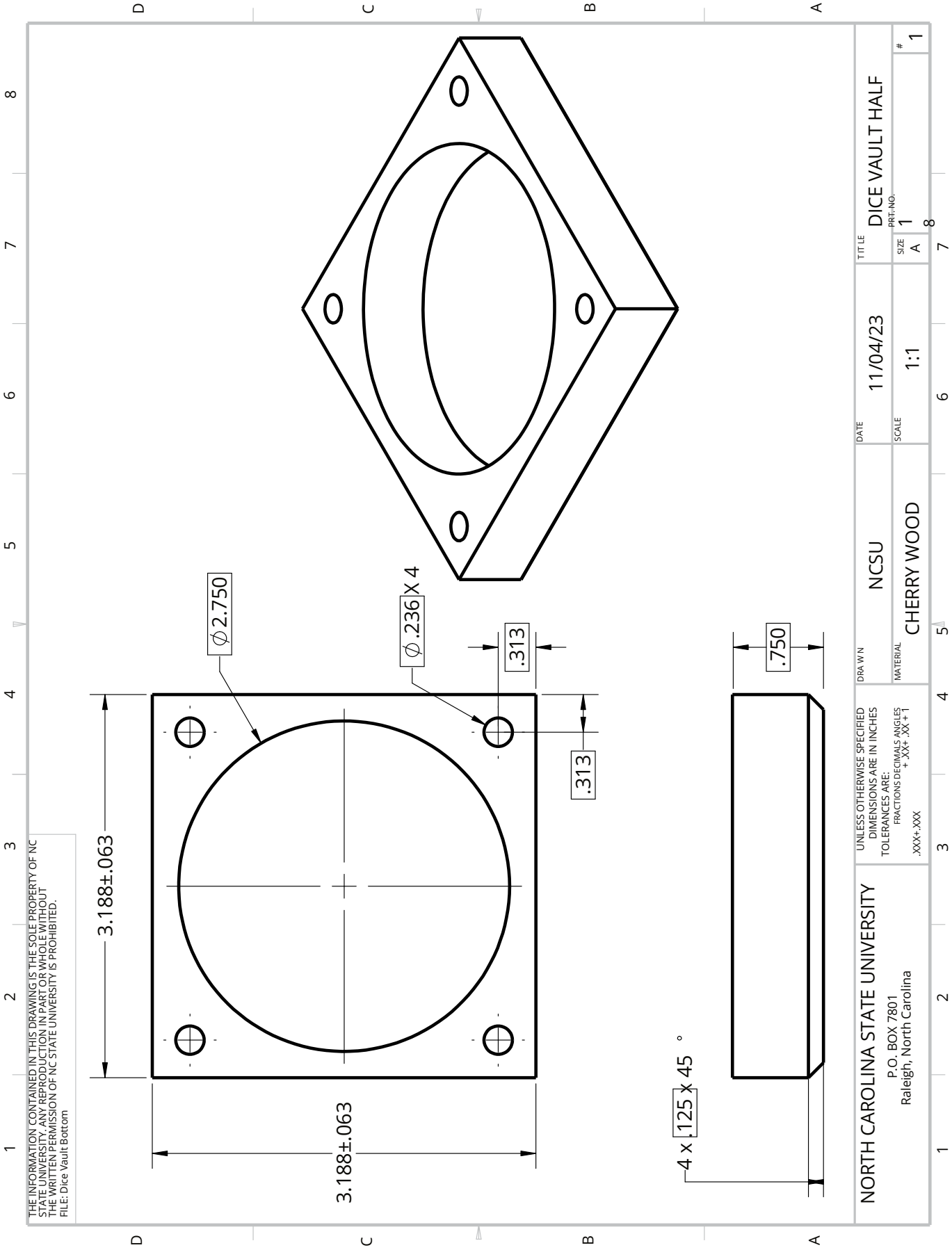


PRODUCT DRAWINGS

3D Isometric View of Dice Vault Half



PRODUCT DRAWINGS - DICE VAULT HALF

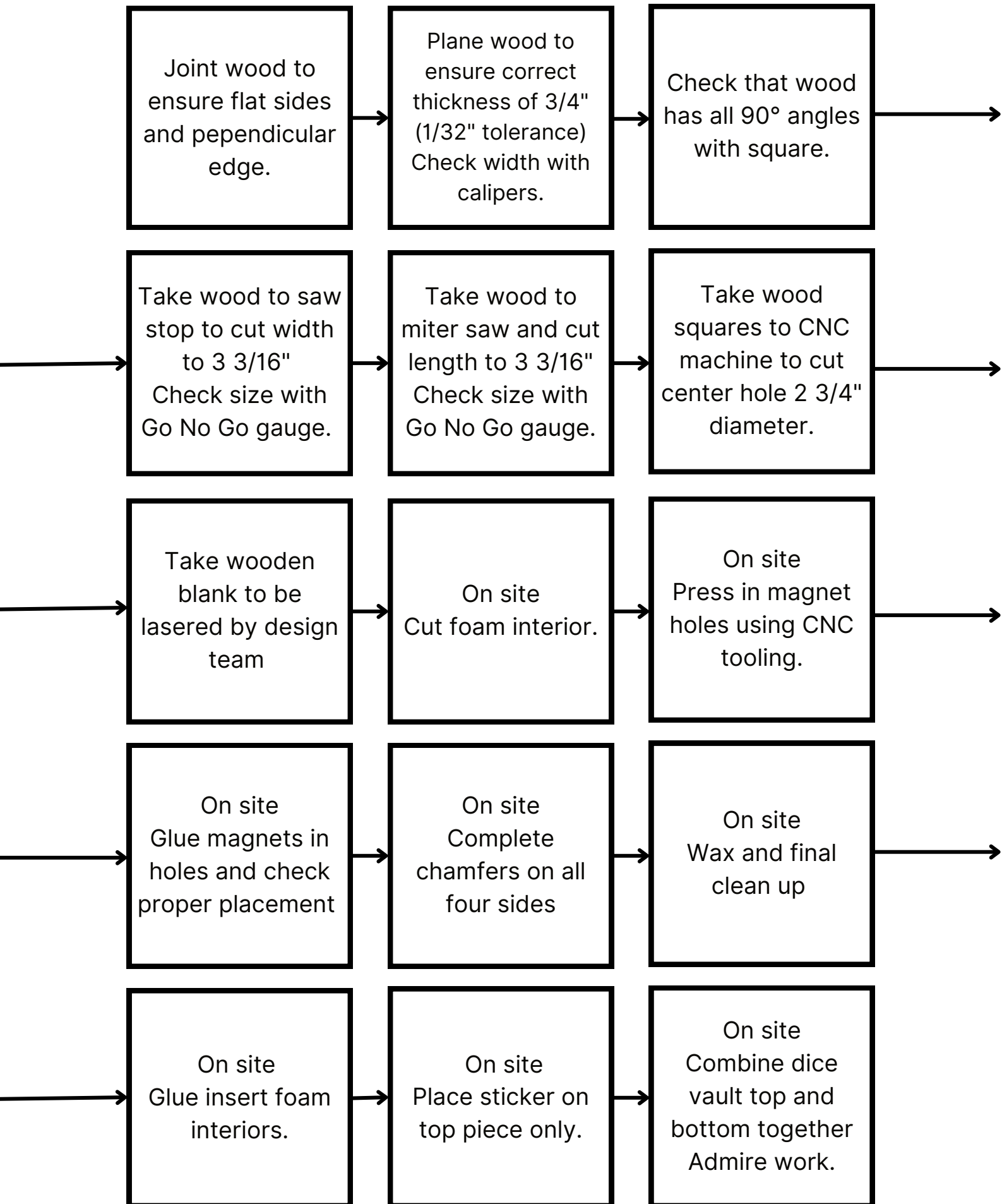


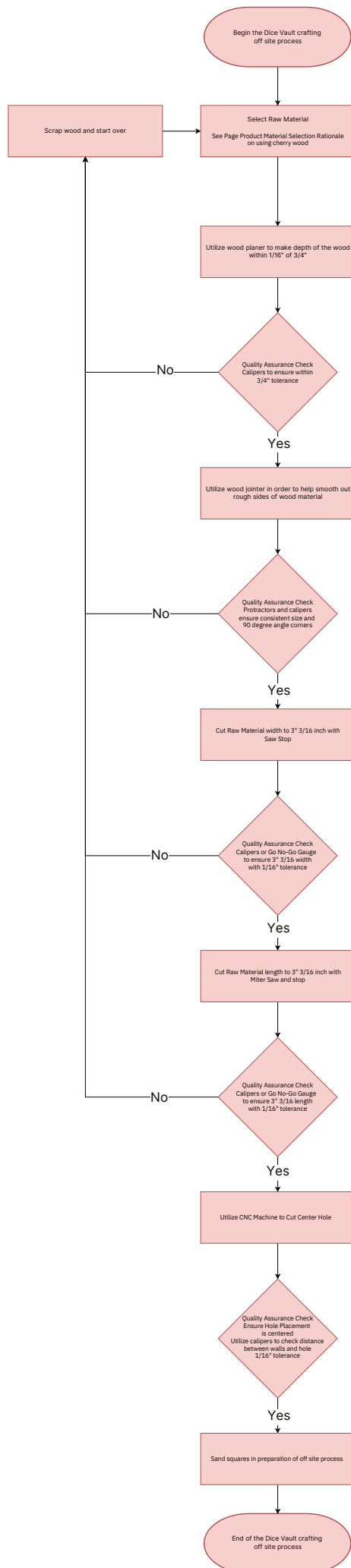
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FILE: Dice Vault Bottom

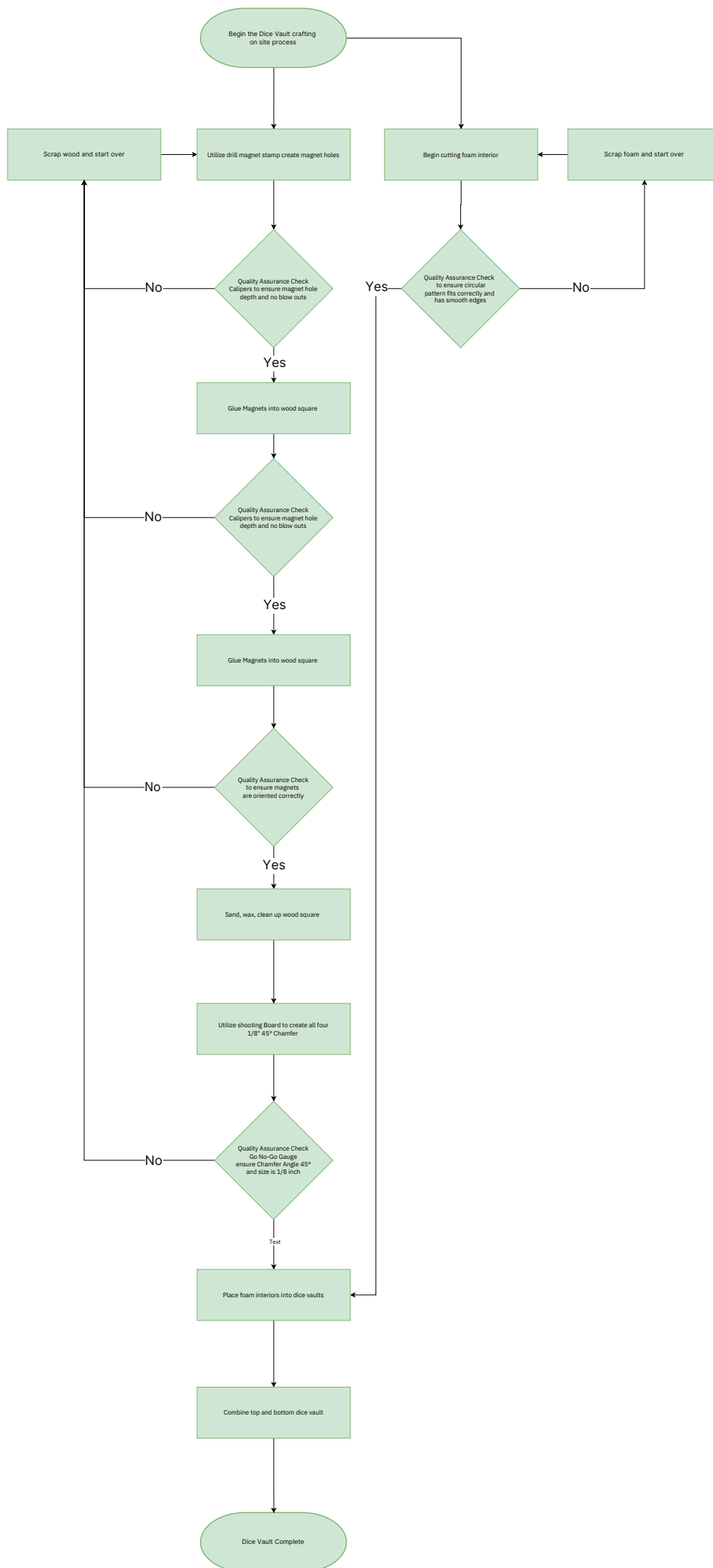
Item No.	Part	MATERIAL	QTY.
1	DICE VAULT HALF	WOOD	1
2	SUPER GLUE	CYANOACRYLATE	2
3	MAGNET	MAGNET	8
4	FOAM INSERTS	SUBLIMATION FOAM	2
5	NC STATE LOGO STICKER	SUBLIMATION PAPER	1

NORTH CAROLINA STATE UNIVERSITY P.O. BOX 7801 Raleigh, North Carolina	UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ARE: FRACTIONS DECIMALS ANGLES " .XX" .XX " ±1 XXX+.XXX	DRAWN	SEAN RYAN	DATE	10/2/2023	TITLE	DICE VAULT	
		MATERIAL	CHERRY WOOD	SCALE	1:2	SIZE	A	
						PRT. NO.	1	
							#	1

OPERATION PROCESS CHART





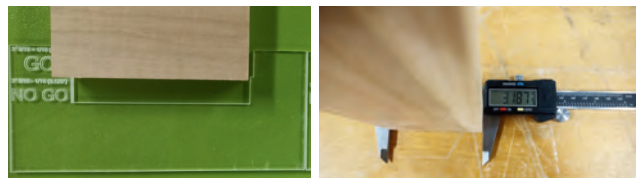


PRE-PRODUCTION PROCESS - THE SAW STOP WOOD CUT

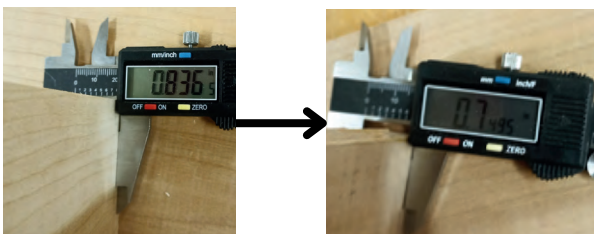
1. Select the desired piece of wood (PRODUCT MATERIAL SELECTION RATIONAL PG #43).
2. Measure to ensure least amount of waste.
3. Utilize Miter Saw if needed to bring length to workable size for the Table Saw.
4. Utilize jointer to flatten out rough face and create perpendicular edge.



5. Utilize the Table Saw to cut piece of wood to appropriate width of $3 \frac{3}{16}$ ". Complete quality assurance check with Go No-Go Gauge or calipers to ensure a tolerance range of $\frac{1}{16}$ " (0.0625).



6. Next utilize a wood planer in order to meet the desired thickness of $\frac{3}{4}$ " (0.75). It is advised to take off $\frac{1}{16}$ " of the wood at a time with half cycle turns. Complete quality assurance check with calipers to ensure a tolerance range of $\frac{1}{16}$ " ($0.0625 = 0.8125 < 0.75 > 0.6875$)



PRE-PRODUCTION PROCESS - THE SAW STOP WOOD CUT

7. Next, the focus will be on cutting the wood to the working length of $3 \frac{3}{16}$ " in order to create a square.

Upon completion of the square. The Go No-Go Gauge should be utilized to ensure that this piece will be usable for later operations. This is especially important as all magnet holes will be created off site at the hotel for the TEECA manufacturing challenge.

A Miter Saw Stop was created (CNC) to ensure accurate cross cuts.

This square will be taken to the Tormach CNC Machine next to have a hole drilled into it.

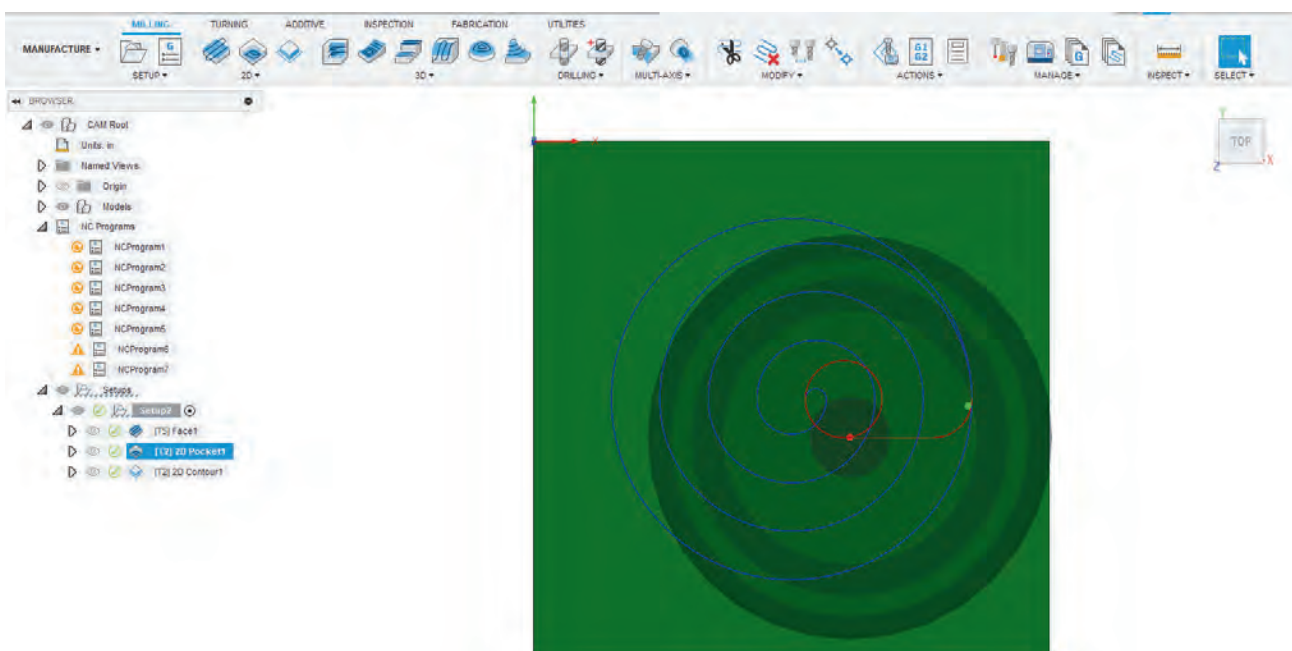
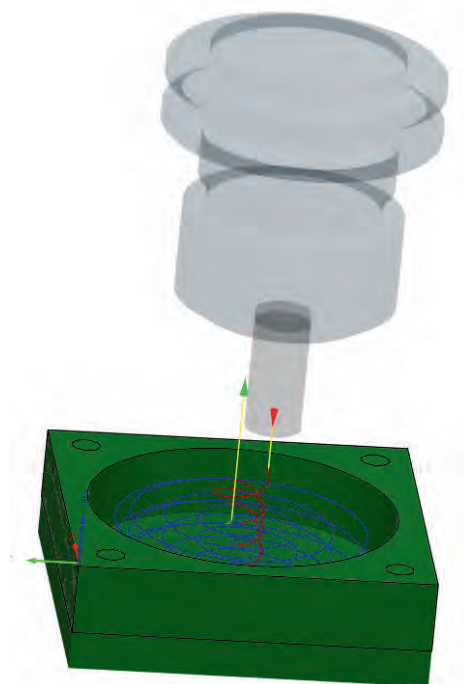
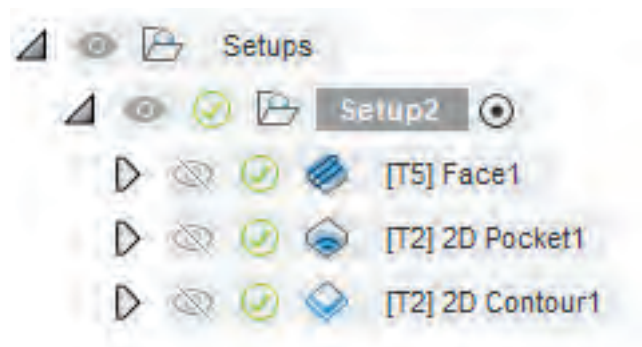
Complete quality assurance check with Go No Go Gauge or calipers to ensure a tolerance range of $\frac{1}{16}$ " (0.0625) from the given expectation of $3 \frac{3}{16}$ ".



PRE-PRODUCTION PROCESS - THE CNC MACHINE AND HOLE CUT

It was determined that Fusion360 was much easier to utilize compared to SolidWorks for CNC tool paths. As such, it was a program that was vital to learn outside of the normal NCSU curriculum for the manufacturing process. The file originally made in SolidWorks was converted to a step file to bring over into Fusion360.

The Fusion360 software creates G-code for the CNC machine to complete two major operations. A 2D pocket and a 2D contour are the two operations to create the center hole. The 2D pocket makes the actual center hole while the contour is a final clean up sweep to clean up messy debris.

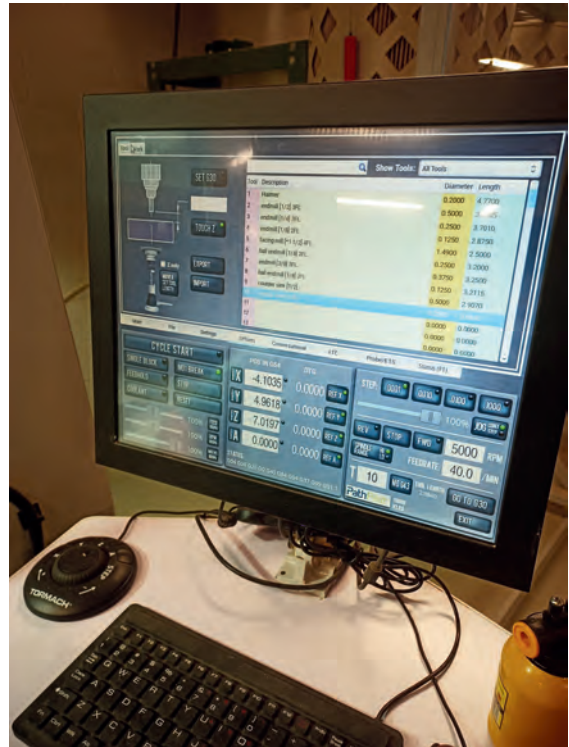


PRE-PRODUCTION PROCESS - THE CNC MACHINE AND HOLE CUT

Transfer and load the Fusion360 file to the Tormach machine by USB.

The Tormach operating system was programmed to repeat the same operation in cutting a center hole on a 3 3/16" wood square.

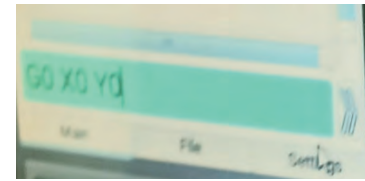
Notes: Profiles saved on G55 and G54 will be utilized for constant changes and projects related outside of the TEECA event.



Sit piece of wood correctly in CNC machine.

Utilize concussive maintenance if needed.

PRE-PRODUCTION PROCESS - THE CNC MACHINE AND HOLE CUT

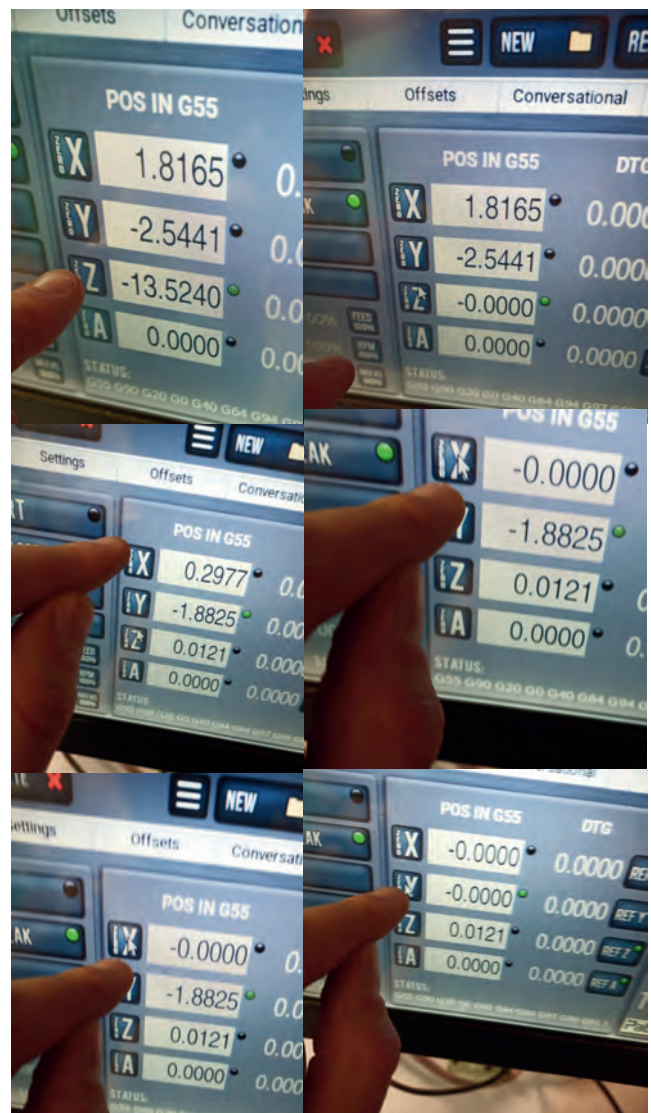


Utilize the command G0 X0 Y0 within the Tormach software in order to ensure that WCS (World Coordinate Systems) work coordinate system is correctly set to the top left hand of the cut square.

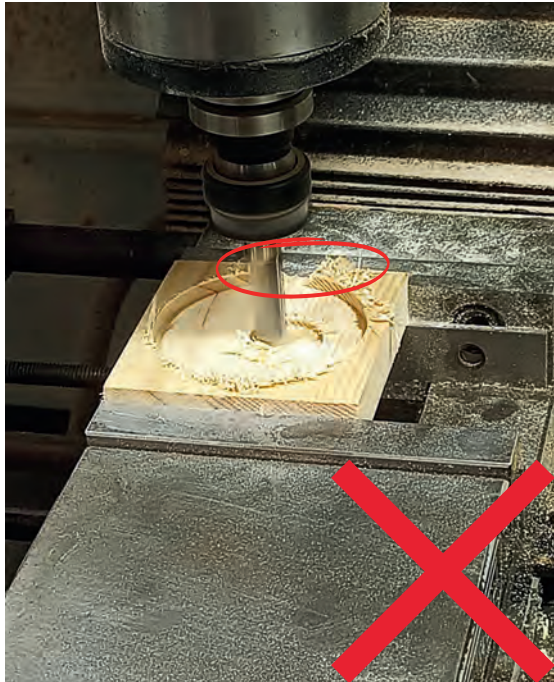
To ensure proper use make sure the correct work offset is selected on the tormach UI. This should only have to be one time but given this is a classroom environment its not unusual for unwanted mistakes or changes to happen.

Record use make sure the correct work offset is selected on the tormach UI

After confirmation that the file is loaded and all axes (x,y,z) are set, you may begin by clicking "Cycle Start".



PRE-PRODUCTION PROCESS - THE CNC MACHINE AND HOLE CUT



NOTE: PLACE GRAIN
HORIZONTAL INTO VICE TO
AVOID BLOW OUTS

Place all wood so that grain is
horizontal to the vice. This will
help avoid blow outs.

The Tormach is an impressive machine that can also be very dangerous and expensive. Observe all cuts to ensure that machine is operating correctly. Be ready to hit the **emergency stop** button if the machine becomes sentient and does something outside of desired actions.



Quality assurance check.

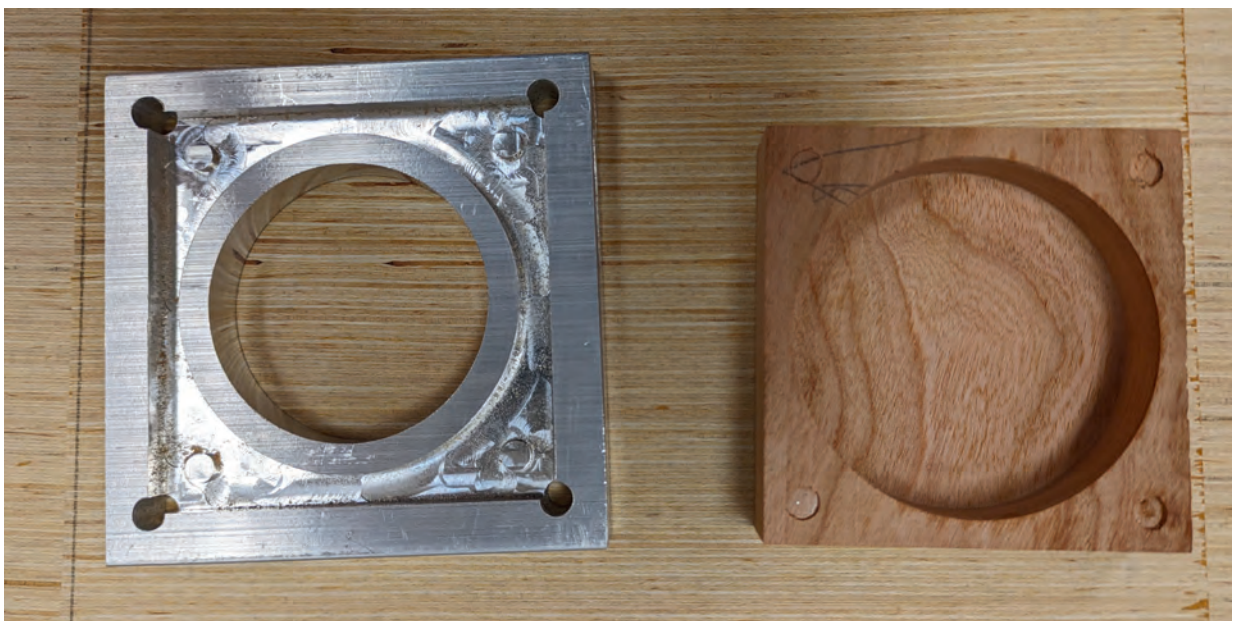
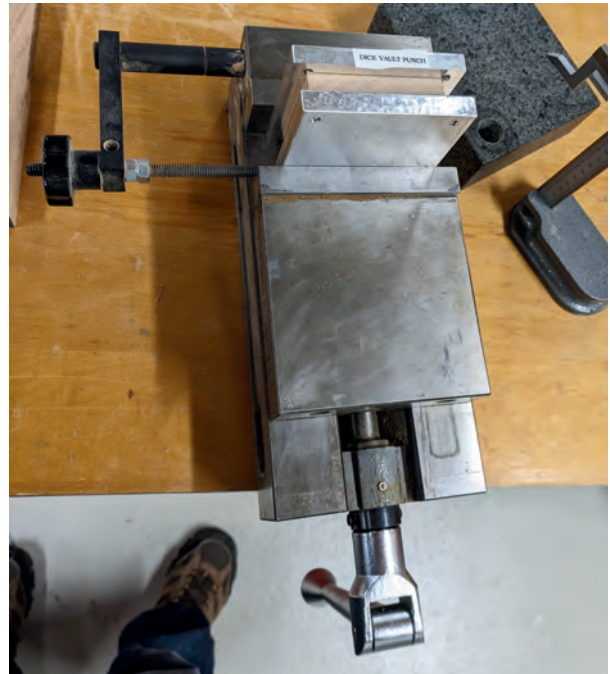
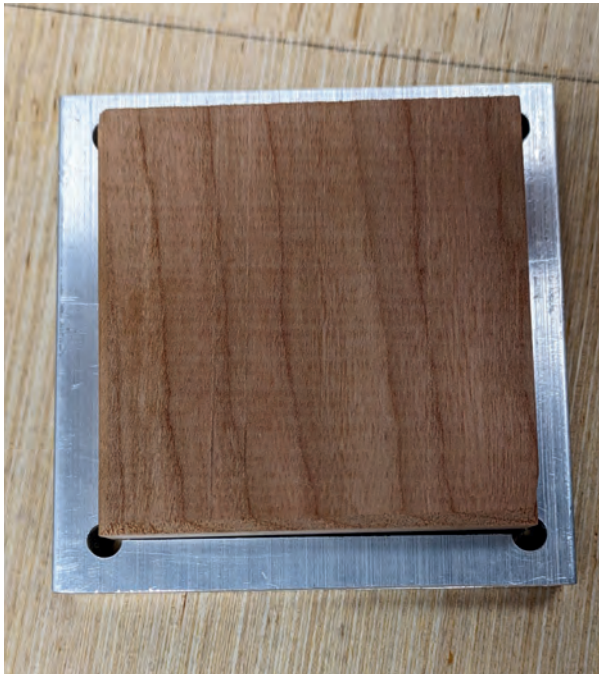
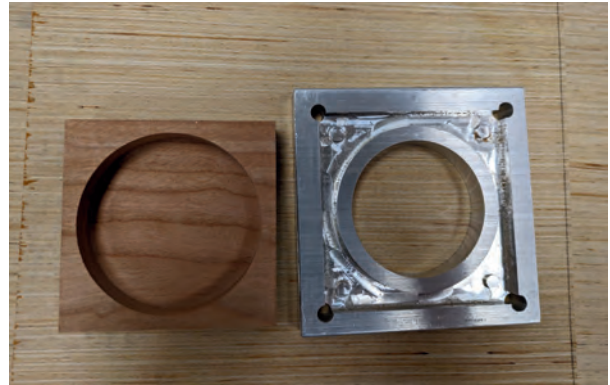
The center hole should fit its
name and visibly be centered.

Confirm this by checking all
sides with calipers to ensure
that all walls are within 1/16"
from the center hole.



PRE-PRODUCTION PROCESS - THE MAGNET HOLE STAMP

Utilize the magnet hole drill punch
with a vice to apply even,
consistent pressure.
Sanding may have to be done to
help smooth out imperfections.

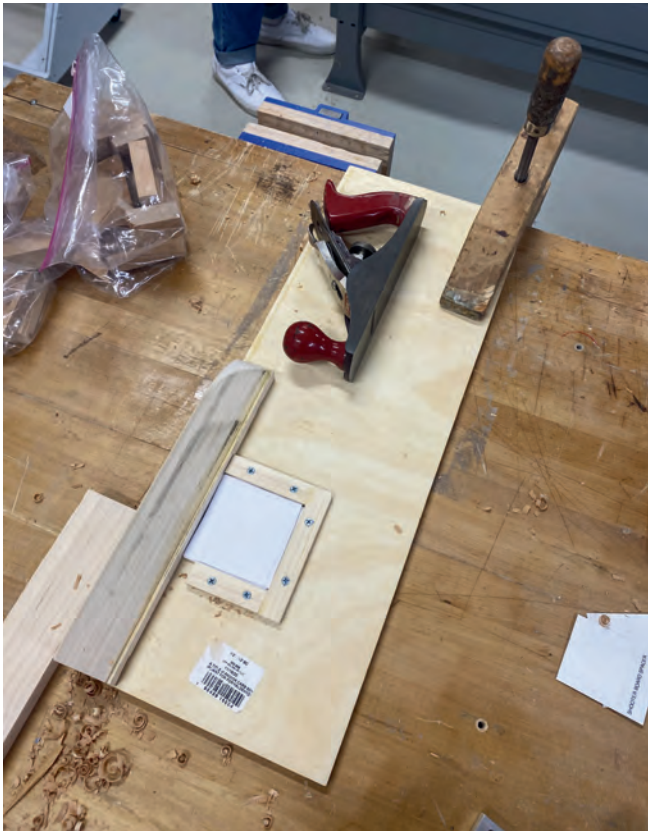


PRE-PRODUCTION PROCESS - THE MAGNET HOLE STAMP

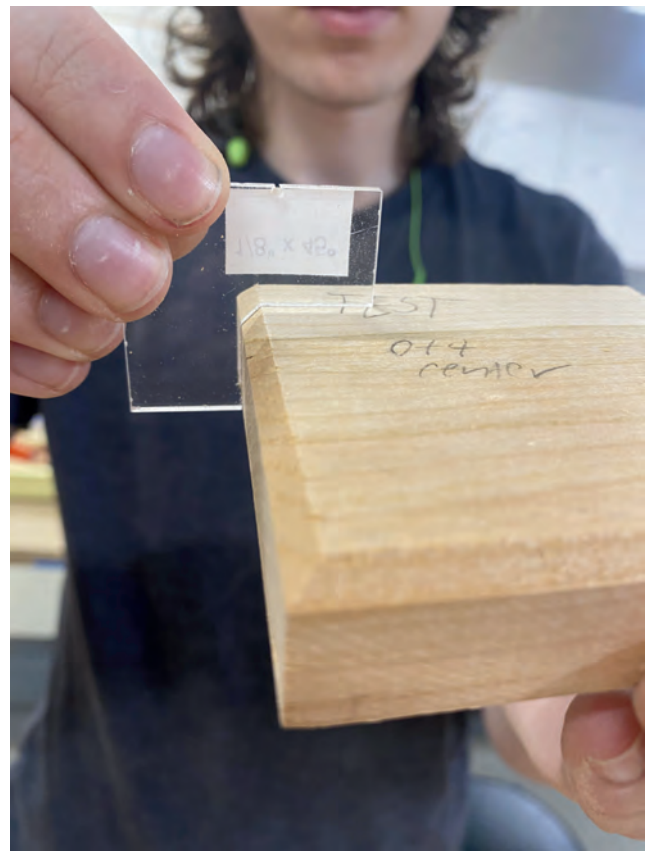
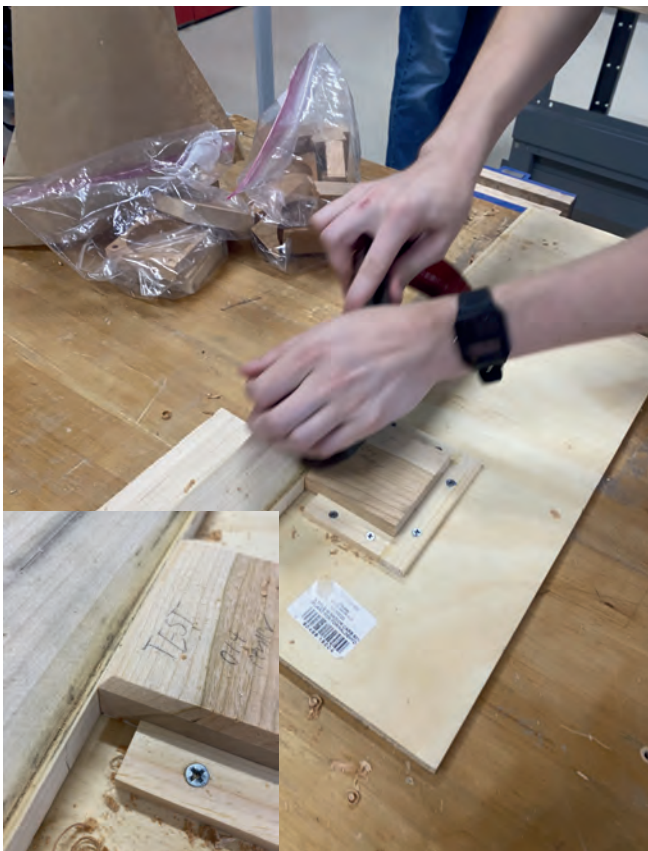
To glue magnets into the dice vault, use the brush of the dual-applicator glue to apply the glue to the magnet holes of both sides. This allows for quick and complete coverage. Once the holes have glue applied, insert two connected magnets into each magnet hole on the bottom half of the vault. Inserting connected magnets ensures that magnets are in correct orientation. Lining up the holes in the top and bottom halves, press the halves together so the glue will adhere the magnets to their respective holes. After 15 seconds, slide the halves apart so the magnets do not pull one another out of their holes. ensure that dice vault halves line up to for a clean look.



PRE-PRODUCTION PROCESS - THE SHOOTING BOARD AND CHAMFER CREATION



Next the shooting board will be utilized to create 45° $1/8"$ chamfer on edges of dice vault. A Go No Go Gauge is utilized to confirm proper cut size and angle has been made.



PRE-PRODUCTION PROCESS - THE LOGO DESIGN & FOAM PLACEMENT

The foam inserts design had an additional design added of the NCSU logo for interior flair. This design was printed on sublimation paper in order to be heat pressed on to the foam.



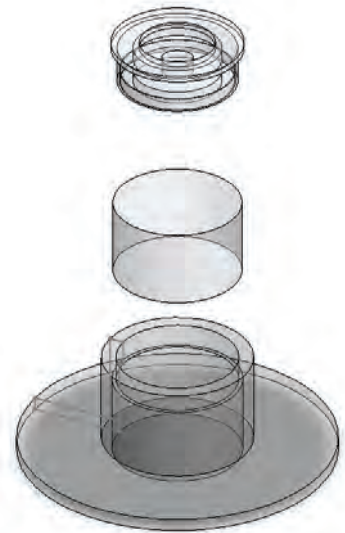
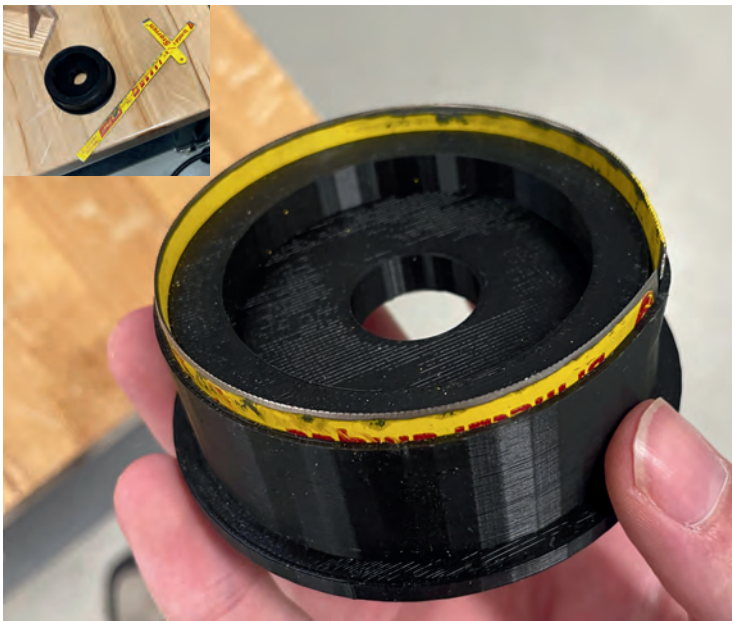
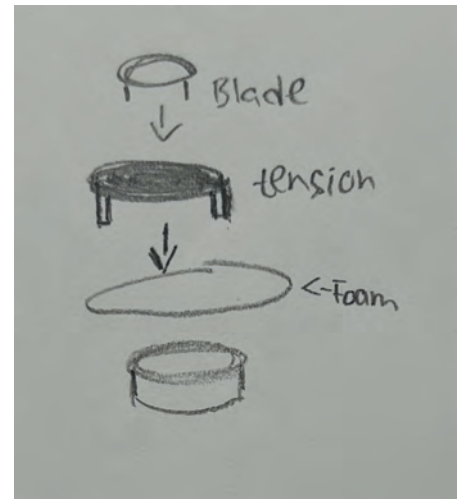
The design is printed and prepped in order to be heated and printed to sublimation foam

A stack of at least 26 pieces (with at least 10 extras) of foam prepped to be utilized on TEECA east conference site



PRE-PRODUCTION PROCESS - THE LOGO DESIGN & FOAM PLACEMENT

A 3D printed tool was created and utilized with a saw blade bent into a circular shape in order to assure accuracy and repetition on the foam pattern.



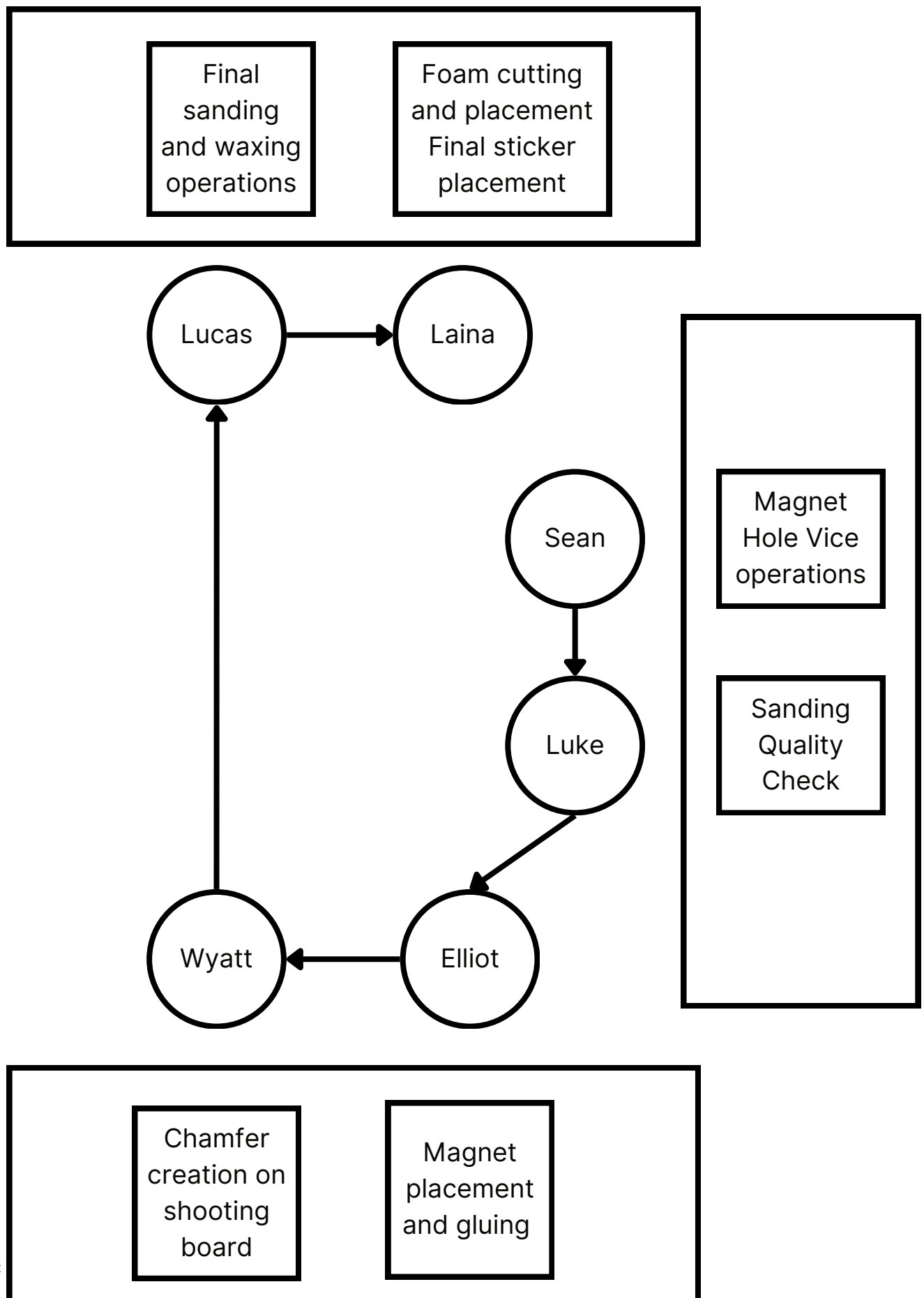
THE COMPLETE PRODUCT

Sanded and waxed for a polished look.



Architectural floor plan of the updated classrooms and labs. The plan shows a central corridor (Room 103) flanked by various rooms. On the left side, from top to bottom, are: a room with a glass wall and flatscreen (Room 117), a room with a glass wall and flatscreen (Room 102), a 3D print room (Room 102), a 3D print space (Room 102), a new fire damper (Room 102), a new fire damper (Room 102), a new fire damper (Room 102), a new fire damper (Room 102), a new fire damper (Room 102), a new fire damper (Room 102). On the right side, from top to bottom, are: a technology lab (Room 104), a robotics lab (Room 104), a TA's office (Room 104B), a fire exit (Room 106), a fire exit (Room 106). The plan includes numerous dimensions, room numbers, and labels for various features like glass walls, flatscreens, and fire exits. A scale of 1/4"=1'-0" is provided at the bottom right.

FACILITIES/PLANT LAYOUT CONFERENCE

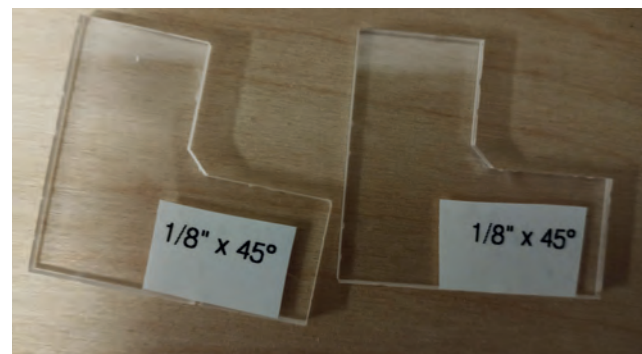


QUALITY CONTROL

Acrylic Go No Go Gauge is created to ensure that rectangle square is appropriate size and within tolerance of $\frac{1}{16}$ " of $3\frac{1}{16}$ "



Acrylic Go No Go Gauge is utilized to confirm that all chamfer cuts are at the expected angle and depth of $\frac{1}{8}$ " x 45°



Calipers a precision measuring instruments. Utilize at multiple points to ensure correct measurements.



PRODUCT MATERIAL SELECTION RATIONAL

Cherry wood emerged as the ideal choice for this dice vault project due to its combination of qualities. First its natural beauty, characterized by a rich, reddish-brown hue and intricate grain patterns. This aesthetic appeal adds a touch of elegance and warmth to the vault. Cherry wood boasts exceptional durability and strength, ensuring the longevity of the finished product. Its robust nature guarantees that the dice vault will withstand years of use without showing signs of wear. Working with cherry wood also has an ideal ratio of ease to work with while still having a high-quality look. This allowed for enhancing the overall design and creating a smooth time line for creating the dice vault.



FINANCIAL INFORMATION

Bill of Materials					
Per Unit					
Part #	Name/Description	Quantity	Dimensions	Material	Description
1	Dice Vault Half Top	1	3" 3/16 Length and Width, 3/4" Height	Cherry wood	Consisten color, clean grain, minimal defects, rough cut
2	Dice Vault Half Bottom	1	3" 3/16 Length and Width, 3/4" Height	Cherry wood	Consisten color, clean grain, minimal defects, rough cut
3	Magnets	8	6mm Diameter, 2mm Height	Magnet	Neodymium Rare Earth Cylinder Magnets
4	Super Glue	1	1 Bottle	Super Glue	Gorilla Super Glue with Brush & Nozzle Applicator
5	Sandpaper	1	1 pack	120 Grit	Utilize for finishing operations
6	Sublimation paper	1	20" x 36"	Sublimation Paper	NCSU logo for top portion of Dice Vault
7	Wax	1	1 container	Finishing Wax	Smooth finish, Clean Appearance
8	Foam	2	2 3/4"	Foam	TahoeBay Blank Beer Can Coolers Sublimation Foam
9	Dice	5	16 mm square	Plastic	6 Sided Standard Colored Dices
10	Game Instructions	1	Paper cut in circle 2 3/4"	Paper	Paper cut in to shape to fit dice vault

FINANCIAL INFORMATION

Per Unit Cost (23 complete Units)	Quantity	Cost per unit	Total Cost
Cherry wood per BF S2S	10	\$2.99	\$29.90
VNDUEEY 300Pcs 6x2mm Refrigerator Magnets	1	\$9.99	\$9.99
Dice (100)	1	\$7.99	\$7.99
Super glue (3 Oz)	1	\$3.99	\$3.99
Foam sheet (TahoeBay Blank Beer Can Coolers (25-Pack) cut in half = 50 pieces for interior)	1	\$15.99	\$15.99
Sand paper	1	\$8.81	\$8.81
Finishing wax	1	\$16.53	\$16.53
Dice	5	\$0.08	\$0.40
Printer paper (game instructions)	1	\$0.01	\$0.01
Tooling Cost			
2 Pieces 1/2" X 4" Aluminum 6061 T6511	1	\$22.79	\$22.79
PLA Filament 2.85	1	\$29.25	\$29.25
Hack saw blade	1	\$5.95	\$5.95
Cherry wood per BF S2S	1	\$2.99	\$2.99
Misc screws	4	Donated	\$0.00
Total Dice Vaults created break even point	50	\$3.02	\$154.59
Recommended sell point 3 times break even point		\$9.28	\$463.78

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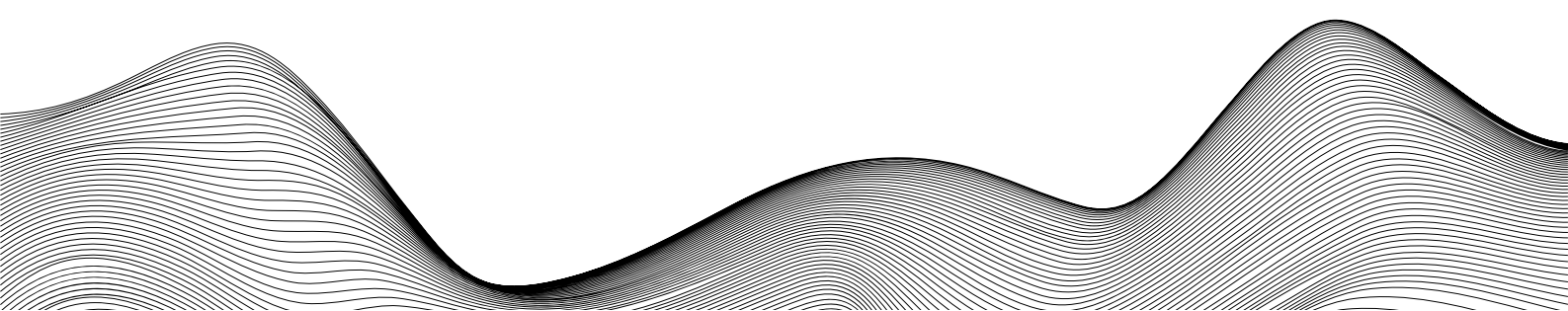
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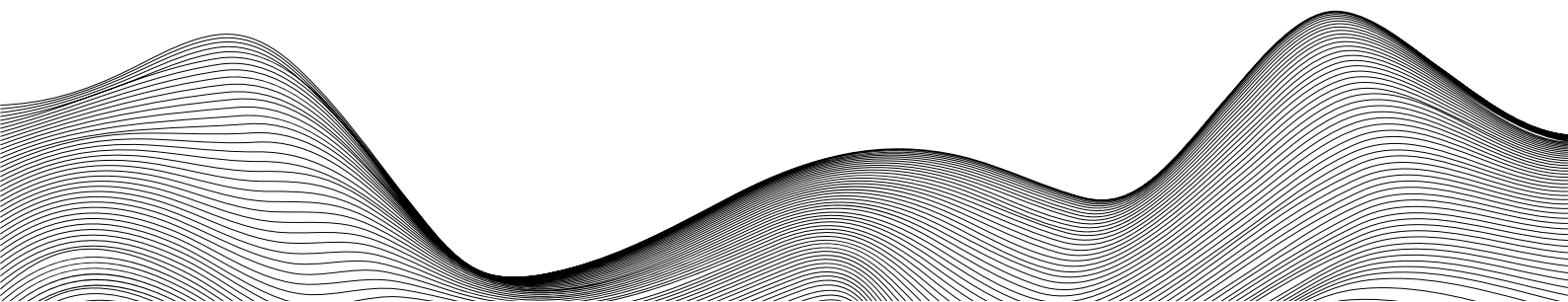
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PRE-PRODUCTION PROCESS - THE SAW STOP WOOD CUT

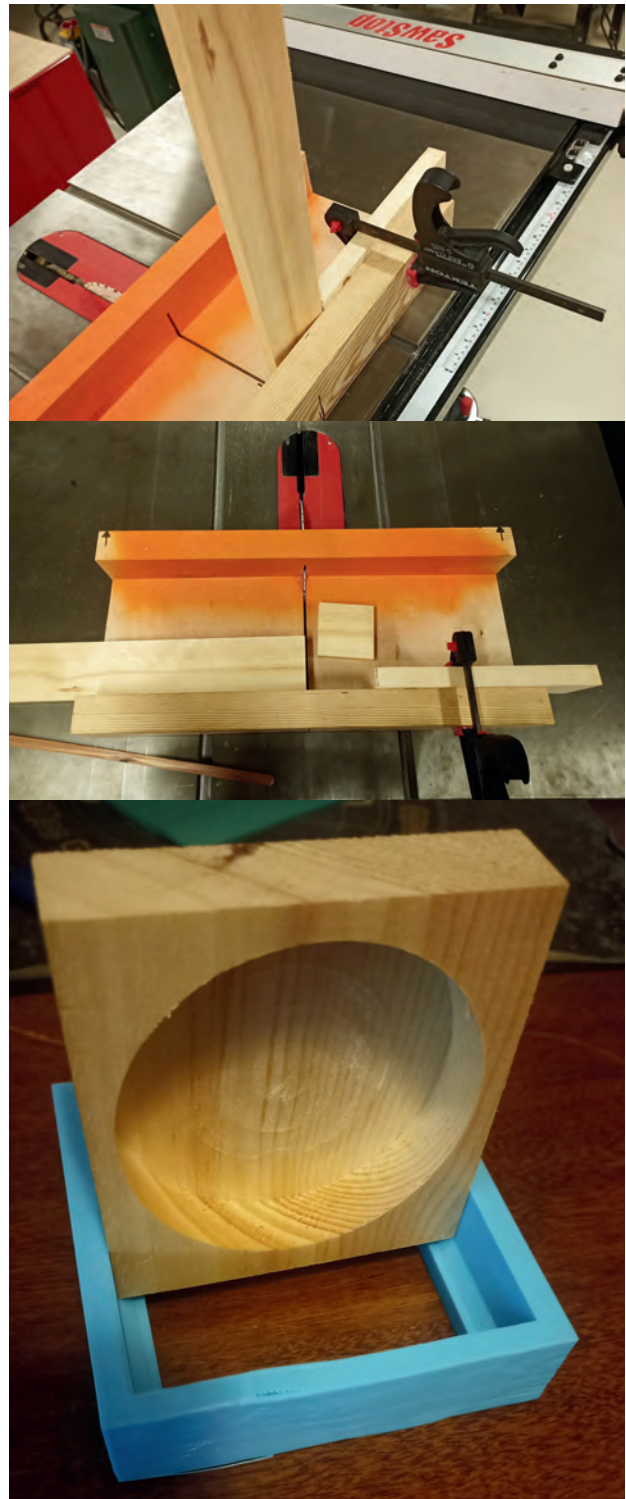
Next, the focus will be on cutting the wood to the width of 3 3/16" in order to create a square.

Utilizing wood that's been cut to 3 3/16" to create a gap of the same size.

Upon completion of the square, the drill hole fixture should be utilized to ensure that this piece will be usable for later operations.

This is especially important as all drill holes will be conducted off site at at the hotel for the TEECA manufacturing challenge.

Next, this square will be taken to the Tormach CNC Machine to have a hole drilled into it. Complete quality assurance check with calipers or drill hole fixture to ensure a tolerance range of 1/16" (0.0625 mm).

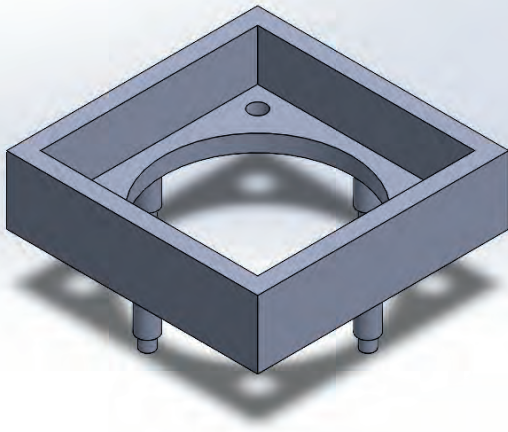
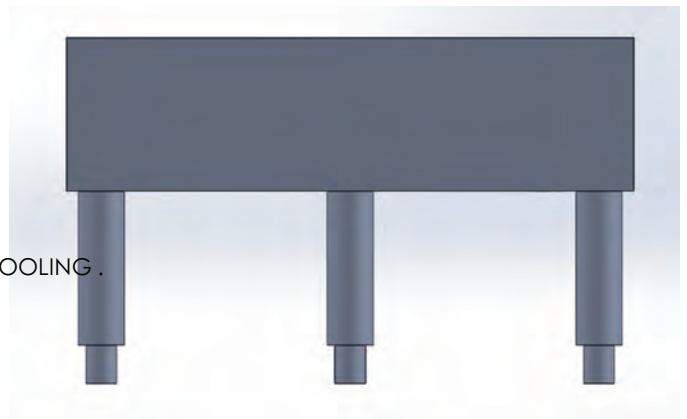


OUTDATED/BACKUP TOOLING .

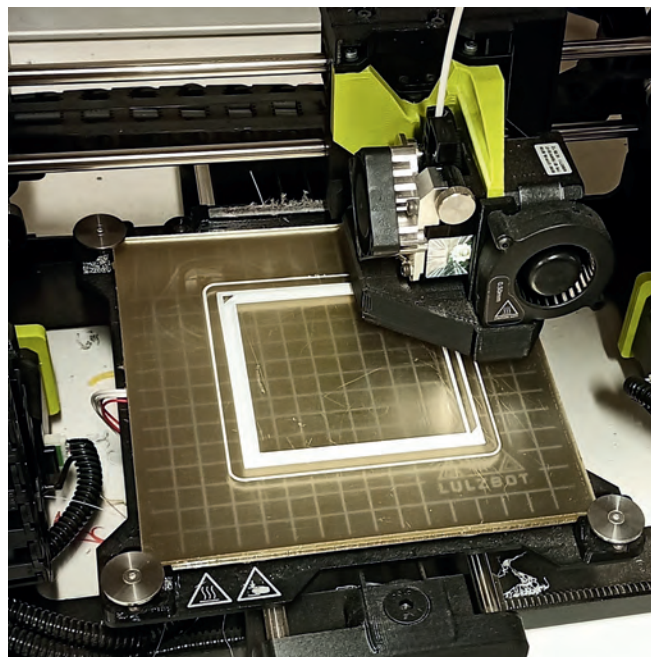
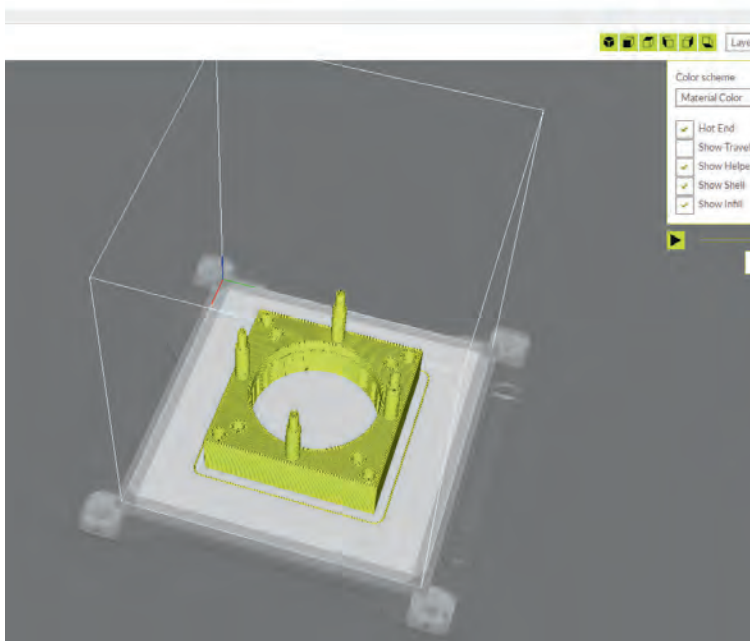
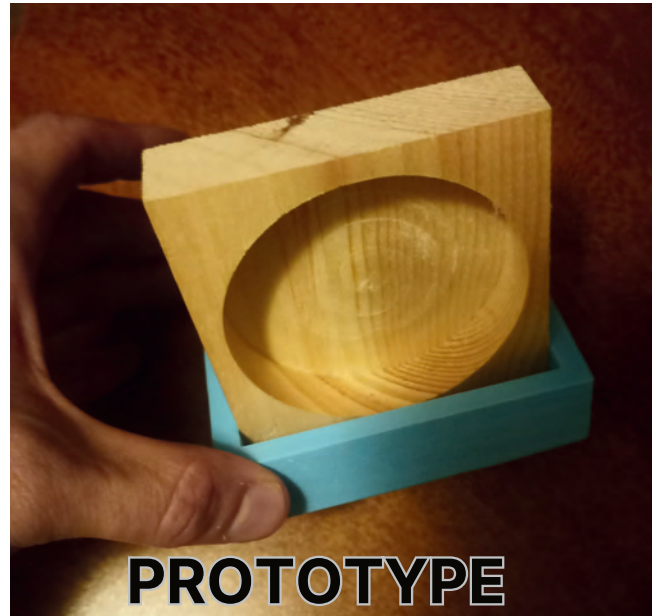
PRE-PRODUCTION PROCESS - DRILL HOLE JIG AND MAGNET PLACEMENT

Jig being created in order to utilize as a Go No Go gauge in terms of size of the Dice Vault Square. Also has an additional use with metals plate in order to properly line up drill holes for magnet placement.

OUTDATED/BACKUP TOOLING .



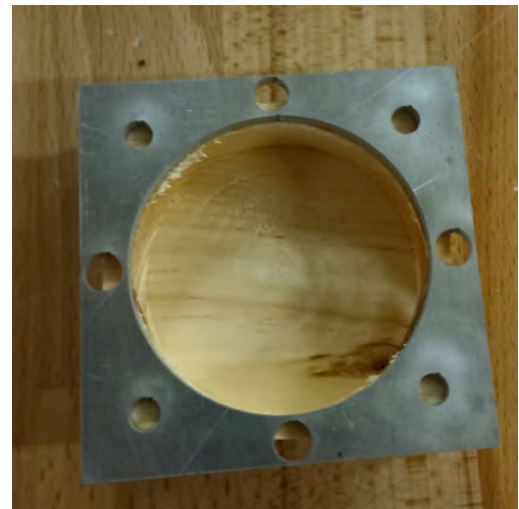
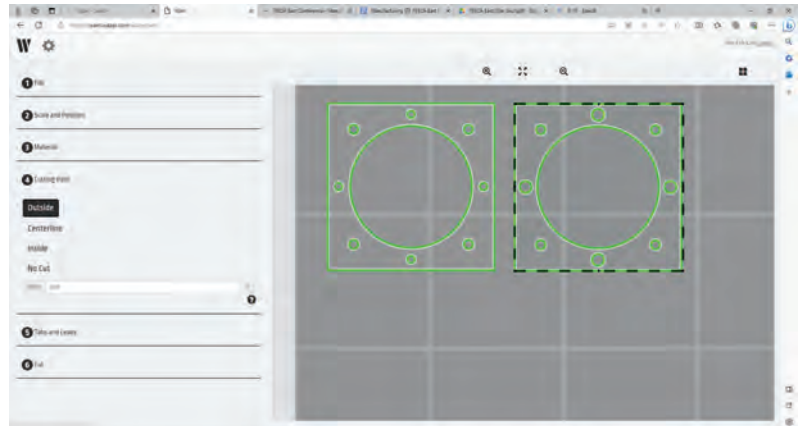
Initial prototype acted as a Go No Go gauge to check the size of the desired square 3" 3/16 on all sides. It was determined we could get more out of this gauge tool by having it act as a tool to line up drill holes. Relief holes were also added to later designs



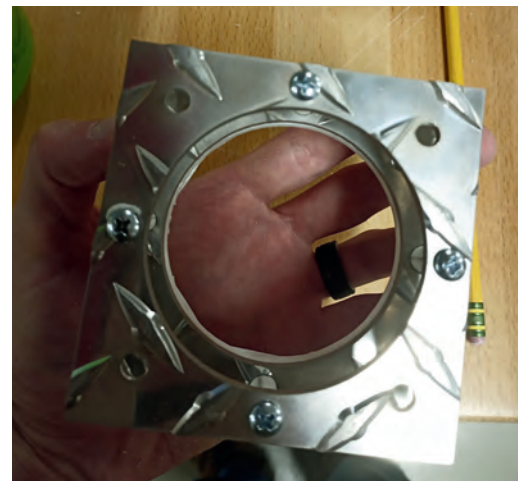
OUTDATED/BACKUP TOOLING.

PRE-PRODUCTION PROCESS - DRILL HOLE JIG AND MAGNET PLACEMENT

Metal face plate created utilizing a WAZER as a component of the dice vault jig in order to line up drill holes correctly in a 90 perpendicular manner. Provides additional protection and stability to 3D printed object.



Metal face plate is adhered to 3D print utilizing screws.



Drill bushings as an additional component where considered. The cost per unit exceeded there lack of use in the final design.



OUTDATED/BACKUP TOOLING.